HYPOCOERCIVE ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We study convergence to non-equilibrium steady states (NESS) for dissipative evolutions on a Hilbert space generated by $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, where $\gamma > 0$ is a friction parameter, $\mathcal{R} \leq 0$ is a self-adjoint dissipator, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta \geq 0$ measures the departure from detailed balance. The central question is: when does the hypocoercive acceleration survive, and how does the driving strength η degrade the convergence rate?

Within the Dolbeault–Mouhot–Schmeiser (DMS) hypocoercivity framework [7, 6], we construct a modified Lyapunov functional and derive an explicit η-dependent convergence rate together with a computable stability threshold $\eta^*(\gamma)$. At the optimal friction $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$—where $\lambda(Q_O)$ is the spectral gap of the self-adjoint reference effective generator—the threshold satisfies $\eta^* \asymp \sqrt{\lambda(Q_O)}$ and the convergence rate improves from the overdamped diffusive scale $\mathcal{O}(\lambda(Q_O)/\gamma)$ to the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$. A complementary overdamped ceiling confirms that this acceleration is inherently a finite-friction phenomenon. Systems with a larger reversible backbone therefore tolerate stronger non-equilibrium driving while retaining the accelerated rate. Applications are given to forced underdamped Langevin dynamics and to a boundary-driven quantum toy model.

1. Introduction

1.1. **Motivation and main question.** Many physical systems of interest—forced diffusions [11, 20], boundary-driven quantum chains [16], active matter models [12]—operate away from detailed balance: their generators contain a non-reversible component of strength η that drives the system to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), it is well known that embedding the dynamics into an enlarged phase space (a *lifting*) can produce a hypocoercive acceleration [5, 19, 8]: at an optimal intermediate friction, the convergence rate improves from the diffusive scale set by the spectral gap of the effective slow dynamics to the square root of that gap—a quadratic speedup. The central question addressed in this paper is:

> For a system that is η away from equilibrium, under what conditions does the hypocoercive acceleration persist, and how does the convergence rate degrade as a function of η?

Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is a friction (scale-separation) parameter, $\mathcal{R} \leq 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ is a non-reversible perturbation whose strength $\eta \geq 0$ is *given by the physics*. In the singular limit $\gamma \to \infty$, adiabatic elimination [14, 25] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. This overdamped limit yields a decay rate of order $\lambda(\mathcal{L}_O)/\gamma$—a diffusive bottleneck. Two

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questions arise: (i) whether the quadratic acceleration at finite friction survives the non-equilibrium perturbation, and (ii) how large η can be before the accelerated regime breaks down.

1.2. **Main results.** The decomposition (1) isolates a self-adjoint *reference effective generator*

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S, \tag{2}$$

which captures the reversible skeleton of the slow dynamics. Let $\lambda(Q_O)$ denote its spectral gap on the mean-zero slow subspace.

Within the DMS hypocoercivity framework [7, 6], we construct a modified Lyapunov functional $\mathcal{E}_{\alpha,\mu}(f) = \frac{1}{2}\|f\|^2 + \alpha\, \Re\langle \mathcal{A}_\mu f, f\rangle$ with an explicit corrector operator $\mathcal{A}_\mu$ adapted from [11, 2], and establish the following.

Theorem A (Theorem 4.1). *Under Assumptions 3.1–3.3, for every $\gamma > 0$ and $\eta \le \eta^*(\gamma)$,*

$$\|e^{t\mathcal{L}} f\| \le \sqrt{3}\, e^{-\lambda_\eta t} \|f\|, \qquad \forall\, f \in \mathcal{H}_0,\ t \ge 0,$$

with an explicit rate $\lambda_\eta \ge \frac{2}{3}(\Lambda_{\mathrm{hypo}}(\gamma) - \eta B_{\mathrm{pert}}(\gamma))$. Here $\Lambda_{\mathrm{hypo}}(\gamma)$ is the unperturbed hypocoercive rate, determined by the spectral gap $\lambda(Q_O)$, the fast dissipation gap λ_R, and the friction γ (see (38) *for the explicit formula), while $B_{\mathrm{pert}}(\gamma)$ quantifies the destabilising effect of the non-equilibrium driving (see* (39)*). The stability threshold $\eta^*(\gamma) := \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ is fully explicit and computable.*

Theorem B (Theorem 4.4). *Under the same assumptions, any admissible L^2 exponential rate ν with prefactor C satisfies the overdamped ceiling*

$$\nu \le C\left[\frac{(1+\kappa_{\mathrm{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1{+}b_2) + \eta(a_1{+}a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

In particular, for $\gamma \to \infty$ and fixed η, the rate is at most $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$: the diffusive bottleneck persists, confirming that acceleration is inherently a finite-friction phenomenon.

Together, Theorems A and B delineate a sharp stability picture. At intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, the convergence rate λ_η scales as $\sqrt{\lambda(Q_O)}$—a quadratic speedup over the diffusive rate $\lambda(Q_O)$. This acceleration persists under non-equilibrium driving precisely when $\eta \le \eta^*(\gamma)$; the rate degrades linearly with η, and the diffusive ceiling $\mathcal{O}(\gamma^{-1})$ from Theorem B ensures that no other friction regime can compensate once η exceeds the threshold.

1.3. **Hypocoercive acceleration and non-equilibrium stability.** The principal contribution of this work is twofold. First, the lifting mechanism achieves a *quadratic speedup*: the convergence rate improves from the diffusive scale $\lambda(Q_O)$ to the kinetic scale $\sqrt{\lambda(Q_O)}$ at an optimal finite friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$. In the perturbative regime $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$ (Remark 3.4), this acceleration is substantial—$\sqrt{\lambda(Q_O)} \gg \lambda(Q_O) \approx \lambda(\mathcal{L}_O)$ whenever the spectral gap of the slow dynamics is small. Second, we provide the first explicit characterisation of how non-equilibrium driving affects this hypocoercive acceleration, whereas prior lifting analyses [5, 9, 1, 19] treat only the reversible case ($\eta = 0$).

The convergence rate degrades *linearly* in η, with a slope $B_{\mathrm{pert}}(\gamma)$ that depends on both the friction γ and the structural constants of the perturbation. At the optimal friction, a nontrivial *singularity cancellation* occurs (Corollary 4.2): the apparent $\lambda(Q_O)^{-1/2}$ blow-up in B_{pert} is exactly compensated by the Lyapunov mixing weight, yielding $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. This has a clear physical interpretation: *the faster the reversible backbone mixes, the larger the non-equilibrium driving it can tolerate*, and the quadratic speedup is preserved throughout the admissible regime $\eta \le \eta^*$.

An important subtlety is that the lifted NESS h_η and the overdamped NESS g_η may differ at finite friction: $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$ in general. However, for lifts constructed via the reverse adiabatic elimination (Section 5.1), one can always enforce $P_F \mathcal{V} P_F = 0$ as a free design choice,

eliminating this bias entirely (Proposition 5.1). In this case $\varepsilon_* = 0$ and the full quadratic speedup applies for all target accuracies; a two-phase extension remains available for natural physical models where $P_F \mathcal{V} P_F \neq 0$ (Remark 5.3).

1.4. **Related work.** The hypocoercive method was pioneered by Villani [26] via modified H^1-type functionals. The L^2 approach based on modified Lyapunov functionals was developed by Dolbeault–Mouhot–Schmeiser [7, 6], who introduced the auxiliary corrector operator that restores coercivity on the degenerate subspace; this DMS framework is the methodological foundation of the present work. The adaptation to Langevin dynamics in weighted Sobolev spaces, using resolvent-type correctors, was carried out by Iacobucci–Olla–Stoltz [11] and Cao–Lu–Wang [2].

The lifting perspective—embedding slow reversible dynamics into an enlarged state space with faster mixing—originates in the Markov chain literature with Chen–Lovász–Pak [5] and has been extended to continuous time by Eberle–Lörler [9] and Brigati–Lörler–Wang [1]. The quantum analogue was recently developed by Li–Lu [19]. All of these works assume the target (slow) dynamics is *reversible* (detailed balance). Our framework extends the theory to non-reversible targets via the perturbative decomposition $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$ and the self-adjoint reference operator Q_O.

Quantitative Zeno-limit convergence in the singular regime $\gamma \to \infty$ was recently established by Carlen–Huse–Lebowitz [3] for boundary-driven quantum systems.

1.5. **Outline.** Section 2 sets up the abstract framework and adiabatic elimination. Section 3 formulates the structural assumptions and introduces the DMS corrector. Section 4 proves the main decay-rate bounds, including the stability threshold and singularity cancellation at optimal friction. Section 5 verifies the framework in two concrete models and develops the reverse adiabatic elimination recipe. Section 6 discusses extensions and open problems.

2. Preliminaries

We set up the abstract Hilbert-space framework and the adiabatic-elimination reduction used throughout.

2.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations. In the noncommutative (quantum) setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ on a finite-dimensional space $\mathfrak{h}$, equipped with the KMS inner product $\langle X, Y \rangle_\sigma := \operatorname{tr}(\sigma^{1/2} X^\dagger \sigma^{1/2} Y)$. The generator is then a GKSL Lindbladian $\mathcal{L}(X) = i[H, X] + \sum_\alpha \mathcal{D}^\dagger[L_\alpha]X$. In the classical (PDE) setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ.

2.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Decompose the generator as $\mathcal{L} = \mathcal{S} + \mathcal{A}$ with $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$ (symmetric part) and $\mathcal{A} := \frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ (skew part).

Definition 2.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f \in \mathcal{H}_0 \setminus \{0\}} (-\langle f, \mathcal{S} f \rangle / \|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \operatorname{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\mathcal{A}$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [26, 7]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \leq C e^{-\lambda t} \|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive*.

Lemma 2.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$*. Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive (*$C = 1$*). Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

2.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \le 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = u + v$ with $u = P_S f$ (macroscopic) and $v = P_F f$ (microscopic).

Definition 2.3 (Centered coupling)**.** The interaction satisfies $P_S \mathcal{V} P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [20, 3].

Definition 2.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{3}$$

In the context of lifting, $\mathcal{L}_O$ is the effective "target" dynamics one seeks to accelerate. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau \mathcal{L}_O}$ as $\gamma \to \infty$ (quantitative Zeno-limit error bounds are given in [3]). From the rate perspective, Theorem 4.4 shows that the decay is bounded by a diffusive γ^{-1} ceiling in this overdamped regime, motivating the finite-friction analysis of Section 4.

When the interaction is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}}$ (skew-adjoint part plus perturbation), the centering condition is enforced componentwise: $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$ and $P_S \mathcal{V}_{\mathrm{pert}} P_S = 0$.

3. The non-equilibrium lifting framework

We now formulate the structural hypotheses for quantitative hypocoercivity at finite friction. The framework identifies the analytic ingredients—spectral gap, centering, flux regularity, and perturbation bounds—needed to transfer coercivity from fast variables to slow variables through a DMS-type corrector [7, 6, 11].

3.1. **Structural assumptions.** Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \ge 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{4}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

The following assumptions encode the structural conditions exploited in Section 4: coercivity of the fast operator, centered coupling, control of slow-induced fluxes, and energy-norm control of non-self-adjoint perturbations.

Assumption 3.1 (Structural Conditions on the Non-Equilibrium Generator)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. We assume the following structural conditions:

(1) **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \ge \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{5}$$

In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \ge 0$ and $\|\mathcal{R}^+\| \le \lambda_R^{-1}$ (the bound follows from the spectral theorem: $\mathcal{R}|_{\mathcal{H}_F} \le -\lambda_R$ implies $\|(\mathcal{R}|_{\mathcal{H}_F})^{-1}\| \le \lambda_R^{-1}$, valid for both bounded and unbounded self-adjoint operators without requiring compact resolvent).

(2) **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{6}$$

Since η is a free parameter in our framework, we enforce this componentwise:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{7}$$

This matches the centered coupling condition (Definition 2.3) used to define the effective generator $\mathcal{L}_O$.

(3) **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reference effective generator defined in (16) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{8}$$

In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\text{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on a common invariant core $\mathcal{C}$ for the operators involved (specified in the examples/appendices) and extended to the relevant energy domains by density. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \le M_R \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle, \qquad \forall \psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{9}$$

Since (9) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \ge 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle (-\mathcal{R})h, h \rangle \le M_R \|h\|^2. \tag{10}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\text{ell}} > 0$ such that for all $\psi \in \text{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\text{skew}} P_F \, (-\mathcal{R}^+) \, g_\psi\| \le \kappa_{\text{ell}} \|P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\| = \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{11}$$

(4) **Structural Decomposition of the Perturbation.** The perturbation operator $\mathcal{V}_{\text{pert}}$ admits a decomposition into a skew-symmetric part and a symmetric part, each controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{12}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f \rangle = 0. \tag{13}$$

Moreover, $\mathcal{K}_{\text{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \ge 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{14}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\text{pert}}$ is symmetric and defined at least on the energy domain $\text{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{15}$$

Equivalently, $\mathcal{S}_{\text{pert}}$ is bounded as a map from $(\text{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\text{pert}}$ is often an unbounded multiplier on plain L^2; (15) is the standard replacement allowing it to be controlled by the fast dissipation.)

Under Assumption 3.1, we recall the effective generator from Definition 2.4, $\mathcal{L}_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion via the **reference effective generator** Q_O:

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S. \tag{16}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle \leq 0. \tag{17}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

Remark 3.2 (Well-definedness of Q_O in unbounded settings). In finite dimensions the composition $Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$ is automatic, but in infinite-dimensional kinetic models $\mathcal{V}_{\text{skew}}$ is an unbounded differential operator and one must check that $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi \in \text{dom}(\mathcal{V}_{\text{skew}})$. This is ensured by Assumption 3.1.3, which stipulates that all compositions are interpreted on a common invariant core $\mathcal{C}$ (e.g. Schwartz functions) preserved by all operators involved; in the Langevin example, the pseudoinverse $\mathcal{R}^+$ is smoothing in the momentum variable, so the intermediate output $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi$ falls in the domain of the second application of $\mathcal{V}_{\text{skew}}$, and Q_O extends to a well-defined self-adjoint operator by integration by parts on $\mathcal{C}$.

Assumption 3.3 (Spectral Gap of Effective Dynamics)**.** Let $\mathcal{K} := \ker(Q_O) \subset \mathcal{H}_S$ be the kernel of the reference generator. We assume Q_O admits a **spectral gap** $\lambda(Q_O) > 0$ on the orthogonal complement $\mathcal{K}^\perp := \mathcal{H}_S \ominus \mathcal{K}$. That is, the Poincaré inequality holds:

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O) \|f - P_{\mathcal{K}} f\|^2, \quad \forall f \in \mathcal{H}_S. \tag{18}$$

Consequently, for any shift parameter $\mu > 0$, the operator $(\mu - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\mu - Q_O)^{-1}\| \leq \mu^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\mu - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

Remark 3.4 (Gap equivalence: $\mathcal{L}_O$ vs. Q_O). By the expansion (57),

$$\mathcal{L}_O = Q_O + \eta\, \mathcal{B}_1 + \eta^2\, \mathcal{B}_2, \tag{19}$$

with model-dependent bounded operators $\mathcal{B}_1, \mathcal{B}_2$ on $\mathcal{H}_S$ in the finite-dimensional setting (and relatively bounded perturbations on the common core in the unbounded examples). Hence

$$\|\mathcal{L}_O - Q_O\| \leq c_1 \eta + c_2 \eta^2 \tag{20}$$

for suitable constants $c_1, c_2 > 0$. If the gap $\lambda(Q_O)$ is isolated and η is such that

$$c_1 \eta + c_2 \eta^2 \leq \theta\, \lambda(Q_O), \qquad 0 < \theta < 1, \tag{21}$$

then resolvent perturbation theory for isolated spectral sets [13] gives persistence of the stable gap:

$$(1 - \theta)\lambda(Q_O) \leq \lambda(\mathcal{L}_O) \leq (1 + \theta)\lambda(Q_O). \tag{22}$$

(Here $\lambda(\mathcal{L}_O)$ denotes the gap measured on the stable part when $\mathcal{L}_O$ is non-self-adjoint.) In particular, in the perturbative regime relevant to this paper, $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$, so the convergence estimate $\lambda_\eta \asymp \sqrt{\lambda(Q_O)} \approx \sqrt{\lambda(\mathcal{L}_O)}$ is consistent.

Remark 3.5 (Scope: self-adjointness excludes Liouvillian Exceptional Points). Assumption 3.3 postulates $\lambda(Q_O) > 0$, which in particular requires Q_O to have a complete set of eigenfunctions (no defective spectrum). Liouvillian Exceptional Points (LEPs)—where two eigenvalues coalesce and the generator becomes non-diagonalizable—are therefore excluded by hypothesis. In our Hilbert-space setting this is not an additional restriction: the self-adjointness of Q_O established above already guarantees diagonalizability by the spectral theorem, so genuine LEPs can only arise in the non-self-adjoint target effective generator $\mathcal{L}_O$ [23], not in Q_O.

3.2. **The DMS corrector.** To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. Following the Dolbeault–Mouhot–Schmeiser (DMS) approach [7, 6], extended to L^2 settings by Iacobucci–Olla–Stoltz [11] and Cao–Lu–Wang [2], this operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level.

Definition 3.6 (DMS Corrector)**.** Let $\mu > 0$ be a shift parameter, to be optimized in Section 4; it is distinct from the reference measures μ_0, μ_η appearing in the examples. We define the **corrector** $\mathcal{A}_\mu : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F. \tag{23}$$

The key obstruction to a direct energy estimate is that dissipation is degenerate on $\mathcal{H}_S$: the plain energy $\frac{1}{2}\|f\|^2$ cannot produce a spectral gap when $\ker(\mathcal{R})$ is non-trivial.

Proposition 3.7 (Why the plain energy method fails)**.** *Assume $\eta = 0$ and $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}}$ with $\mathcal{V}_{\text{skew}}$ skew-adjoint and $\ker(\mathcal{R}) = \mathcal{H}_S \neq \{0\}$. Then there is no constant $\lambda > 0$ such that*

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2 \leq -\lambda\,\|f_t\|^2 \qquad \textit{for all solutions } \partial_t f_t = \mathcal{L} f_t. \tag{24}$$

Proof. For any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\mathcal{R}\phi = 0$ and $\Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0$, hence

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2\Big|_{t=0,\, f_0=\phi} = \Re\langle \mathcal{L}\phi, \phi\rangle = \gamma\langle \mathcal{R}\phi, \phi\rangle + \Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0. \tag{25}$$

If such a $\lambda > 0$ existed, we would get $0 \leq -\lambda\|\phi\|^2 < 0$, a contradiction. □

Proposition 3.7 confirms that the energy $\frac{1}{2}\|f\|^2$ alone is degenerate on $\mathcal{H}_S$. The remedy is to augment it with a cross term involving the corrector $\mathcal{A}_\mu$; differentiating $\Re\langle \mathcal{A}_\mu f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\mu$, including the control over non-equilibrium perturbations.

Lemma 3.8 (Structural Estimates for the DMS Corrector)**.** *Let $\mu > 0$ be the resolvent shift parameter. The corrector $\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ satisfies the following structural estimates, where Q_O is the reference effective dynamics.*

Under Assumption 3.1, the operator $\mathcal{A}_\mu$ satisfies the following structural estimates uniformly in μ. The constants defined below depend solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$:

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\mu\| \leq k_{\mathcal{A}}\mu^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{26}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \leq k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{27a}$$

$$\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| \leq k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{27b}$$

The dual estimate (27b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 3.1.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}\| \leq k_P(\mu), \tag{28a}$$

$$\|\mathcal{A}_{\mu}\mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\mu). \tag{28b}$$

with control constants

$$k_P(\mu) := (b_1 + b_2)k_{\mathcal{A}}\,\mu^{-1/2}, \tag{29}$$

$$\tilde{k}_P(\mu) := \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}. \tag{30}$$

Moreover, the centered perturbation coupling in Assumption 3.1.2 implies

$$P_S\mathcal{V}_{\text{pert}}P_S = 0 \qquad \Longrightarrow \qquad P_S\mathcal{V}_{\text{pert}}^{\dagger}P_S = 0, \tag{31}$$

and therefore

$$\mathcal{V}_{\text{pert}}\,\mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^{\dagger}\,\mathcal{H}_S \subset \mathcal{H}_F. \tag{32}$$

Proof. See Appendix B. □

3.3. **The modified Lyapunov functional.** We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the corrector $\mathcal{A}_{\mu}$.

Definition 3.9 (Modified Lyapunov functional [7, 6])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_{\mu}f, f\rangle. \tag{33}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_{\mu}$.

Proposition 3.10 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_{\mu}\| \leq 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_{\mu}\|)\|f\|^2 \leq \mathcal{E}_{\alpha,\mu}(f) \leq \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_{\mu}\|)\|f\|^2. \tag{34}$$

Proof. The proof follows directly from the Cauchy-Schwarz inequality $|\Re\langle\mathcal{A}_{\mu}f, f\rangle| \leq \|\mathcal{A}_{\mu}\|\|f\|^2$ and the triangle inequality. □

4. Convergence bounds for the lifted semigroup

We now derive quantitative decay estimates for the semigroup on the mean-zero subspace. A constructive lower bound (via a 2×2 dissipation matrix) and a complementary overdamped upper ceiling (via singular values) together locate the finite-friction acceleration window.

4.1. **Lower bound via modified hypocoercivity.** This subsection derives an explicit lower bound for the spectral decay rate by jointly optimizing the two Lyapunov parameters (α, μ). The optimization is fully algebraic and yields both the friction-dependent rate profile and a perturbative admissibility threshold.

The complete matrix estimates and parameter optimization are deferred to Appendix C; the explicit formulas used throughout the sequel are stated below.

Notation for the rate functions. The following structural scalars, depending only on the parameters in Assumption 3.1, are fixed throughout this section:

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{35}$$

where $k_{\text{reg}} = \sqrt{M_R/\lambda_R}$ (see (27a)) and κ_{ell} is from (11). Here Θ is the slow–fast flux amplitude, K_{geom} aggregates the geometric coupling constants from the corrector bounds, and $b_{2,\text{eff}}$ incorporates the fast-mode energy amplification $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R \|f_F\|^2$.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{36}$$

which measures the effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{37}$$

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}, \tag{38}$$

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}, \tag{39}$$

where $C_{\text{sing}} := 2(b_1+b_2)\, k_{\mathcal{A}} + \frac{3}{4}(a_1+a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 3.8. Here $\alpha_*(\gamma)$ is the optimal Lyapunov mixing weight, $\Lambda_{\text{hypo}}(\gamma)$ is the friction-dependent lower bound on the decay rate, and $B_{\text{pert}}(\gamma)$ bounds the perturbation's destabilizing effect. The apparent $\lambda(Q_O)^{-1/2}$ singularity in B_{pert} is cancelled at the optimal friction by $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ (Corollary 4.2).

Main result. The structural properties established in Appendix C yield a uniform norm equivalence (Proposition 3.10) with prefactor $C = \sqrt{3}$ for all $\gamma > 0$.

Theorem 4.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 3.1 and 3.3 hold. Set $\eta^*(\gamma) := \Lambda_{\text{hypo}}(\gamma)/(2B_{\text{pert}}(\gamma))$. For any $\gamma > 0$, if $\eta \leq \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on $\mathcal{H}_0$:*

$$\|e^{t\mathcal{L}} f\| \leq \sqrt{3}\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{40}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \geq \frac{2}{3}\left(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\right). \tag{41}$$

Sketch of Proof. The full computation is in Appendix C. With the resolvent shift fixed at $\mu_* = \lambda(Q_O)$, the Lyapunov derivative satisfies $\frac{d}{dt}\mathcal{E}_{\alpha,\mu_*} \leq -\mathbf{v}^T\mathbf{M}(\alpha)\mathbf{v}$ for the vector $\mathbf{v} = (\|P_F f\|, \|P_S f\|)^T$ and a 2×2 positive-definite dissipation matrix $\mathbf{M}(\alpha)$. Maximising over α yields the optimal weight $\alpha_*(\gamma)$ from (37) and the rate $\Lambda_{\text{hypo}}(\gamma)$. The distortion parameter $\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\mu_*}\|$ satisfies

$$\delta(\gamma) \leq \frac{\sqrt{\lambda_R}}{4\Theta K_{\text{geom}}} \leq \frac{1}{2} \qquad \forall\, \gamma > 0, \tag{42}$$

by the solver bound $\|\mathcal{A}_{\mu_*}\| \leq (2\sqrt{\lambda_R})^{-1}\lambda(Q_O)^{-1/2}$, the formula for α_* in (37), and the elementary extremum $\sup_{x>0} x(x+K_{\text{geom}})^{-2} = (4K_{\text{geom}})^{-1}$. This gives the uniform equivalence $\frac{1}{4}\|f\|^2 \leq \mathcal{E}_{\alpha_*,\mu_*} \leq \frac{3}{4}\|f\|^2$, hence the prefactor $\sqrt{3}$ and the $\frac{2}{3}$ factor in (41). □

Regime summary. The formula (38) displays two distinct regimes for the lower bound $\Lambda_{\text{hypo}}(\gamma)$. As $\gamma \to \infty$, the denominator $Z(\gamma)^2 \sim \gamma^2\Theta^2\lambda(Q_O)^{-1}$ grows quadratically, giving diffusive scaling $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$. At small or order-one γ, the denominator is controlled by the γ-independent constants $K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}$, and the bound scales linearly: $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$. The maximum therefore occurs at an intermediate friction, with kinetic-type scaling $\max_\gamma \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

Corollary 4.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 4.1. Assume furthermore that the structural constants* Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\mathrm{eff}}$, *and* C_{sing} *are independent of* $\lambda(Q_O)$ *(in particular, they remain* $\mathcal{O}(1)$ *as* $\lambda(Q_O)\downarrow 0$*), and that* $b_{2,\mathrm{eff}}>0$.

Then the unperturbed rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ *defined in* (38) *attains its global maximum at*

$$\gamma_{\mathrm{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\mathrm{geom}}^2 + \Big(2k_{\mathrm{reg}} + \tfrac{1}{2}\Big)} \asymp \sqrt{\lambda(Q_O)}. \tag{43}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{44}$$

Moreover, the mixing weight in (39) *satisfies*

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{45}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ *in* $B_{\mathrm{pert}}(\gamma)$ *is cancelled at the optimal friction. Consequently,* $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ *and the stability threshold satisfies*

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{46}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\big)^2 + \Big(2k_{\mathrm{reg}} + \frac{1}{2}\Big)}. \tag{47}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\mathrm{reg}} + \frac{1}{2}$. Then $\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\mathrm{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (39), $B_{\mathrm{pert}}(\gamma) = b_{2,\mathrm{eff}} + \alpha_*(\gamma)\,C_{\mathrm{sing}}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\mathrm{opt}}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\mathrm{opt}}) = x + K_{\mathrm{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{Z(\gamma_{\mathrm{opt}})^2+2k_{\mathrm{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$. Finally, $\eta^*(\gamma) = \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ yields $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. □

Define the (rigorous) L^2 relaxation time as in [19] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \ge 0\,;\ \|e^{t\mathcal{L}}f\| \le e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{48}$$

The following holds:

Corollary 4.3 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem 4.1. Then for* $\eta \le \eta^*(\gamma)$*, the exponential decay estimate from Theorem 4.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \le \frac{1+\log\sqrt{3}}{\lambda_\eta}. \tag{49}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ *(Corollary 4.2) and for perturbations within the stability threshold, one has*

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{50}$$

Proof. By Theorem 4.1, for all $f \in \mathcal{H}_0$ and $t \ge 0$ one has $\|e^{t\mathcal{L}}f\| \le \sqrt{3}\,e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1+\log\sqrt{3})/\lambda_\eta$ so that $\sqrt{3}\,e^{-\lambda_\eta t} = e^{-1}$. □

4.2. **Overdamped diffusive ceiling.** To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f \in \mathcal{H}_0 \cap \mathrm{dom}(\mathcal{L}) \setminus \{0\}} \frac{\|\mathcal{L} f\|}{\|f\|}, \tag{51}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversiblizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g., [21]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$ [4].

Theorem 4.4 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumptions 3.1.1–4 and Assumption 3.3. Assume also* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume moreover that* $\lambda(Q_O)$ *is attained by a normalized eigenfunction* $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ *(automatic in finite dimensions, and in many PDE settings with compact resolvent). Let* $C \geq 1$ *and* $\nu > 0$ *be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{52}$$

Then for every $\gamma > 0$,

$$\nu \leq C\, s(\mathcal{L}) \leq C \left[\frac{(1+\kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{53}$$

where κ_{ell} *is the flux-regularity constant from Assumption 3.1.3 and* a_1, a_2, b_1, b_2 *are from Assumption 3.1.4. In particular, for* $\eta \leq c/\gamma$ *one obtains the diffusive overdamped scaling* $\nu = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$.

Sketch of Proof. The proof is given in Appendix D. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$. The flux regularity bound (11) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\mathrm{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption 3.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. □

Theorem 4.4 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem 4.1, this confirms that the accelerated $\sqrt{\lambda(Q_O)}$ regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

Corollary 4.5 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem 4.4. Define* $t_{\mathrm{rel}}(\mathcal{L})$ *as in Corollary 4.3. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \leq \frac{(1+\kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{54}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2, with the same prefactor C *as in Theorem 4.4):*

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2C\, s(\mathcal{L})}. \tag{55}$$

Consequently, if $\eta \leq c/\gamma$, *then* $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$ *and hence*

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{56}$$

In particular, the kinetic scaling $t_{\mathrm{rel}}(\mathcal{L}) \asymp \lambda(Q_O)^{-1/2}$ *is achieved at optimal friction, combined with Corollary 4.3.*

5. Examples

We illustrate the abstract framework through two models. Section 5.1 describes a systematic *reverse adiabatic elimination*: given an effective generator $\mathcal{L}_O$ on the slow space, one constructs a lifted dynamics whose adiabatic elimination recovers $\mathcal{L}_O$ exactly. Section 5.2 treats forced underdamped Langevin dynamics, where the kinetic equation arises as the reverse adiabatic elimination of the overdamped Fokker–Planck operator. Section 5.3 gives a finite-dimensional boundary-driven Lindbladian with non-self-adjoint effective slow dynamics.

5.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** A practitioner typically starts with a *nonequilibrium effective generator* $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$: it encodes the slow dynamics one observes, or wishes to simulate, in its full non-equilibrium form. The goal is to *reverse-engineer* a lifted dynamics $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ on an enlarged space $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ whose adiabatic elimination (Definition 2.4) recovers $\mathcal{L}_O$ exactly. We call this procedure *reverse adiabatic elimination* and describe it in three explicit steps.

Perturbation expansion of $\mathcal{L}_O$. By Definition 2.4, the effective generator satisfies $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. Expanding the product and separating orders of η gives

(57)
$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\,P_S}_{Q_O}\ \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\ +\ \mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)}\ \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}.$$

Here Q_O is the self-adjoint *reversible backbone* (see (16)): it equals the effective generator one would obtain at equilibrium ($\eta = 0$). The remaining terms—the $\mathcal{O}(\eta)$ cross terms and the $\mathcal{O}(\eta^2)$ purely perturbative correction—are bounded operators whose norms enter the effective overdamped gap analysis. A concrete formula for each term appears in (74) for the boundary-driven model.

Step 1: Identify the reversible backbone Q_O. The practitioner decomposes $\mathcal{L}_O$ by setting $\eta = 0$ in the physical model: the resulting self-adjoint generator is Q_O. Its spectral gap $\lambda(Q_O)$ controls both the accelerated convergence rate and the stability threshold $\eta^*(\gamma)$ in Theorem 4.1.

Step 2: Factor Q_O as a Schur complement. The adiabatic-elimination formula (16) shows that Q_O has the algebraic structure of a *Schur complement*:

$$Q_O = -T^*\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\mathrm{skew}}P_S, \tag{58}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. Reversing this: given Q_O, choose an auxiliary Hilbert space $\mathcal{H}_F$ and a fast dissipator $\mathcal{R}_F < 0$ on $\mathcal{H}_F$, then find a coupling $T : \mathcal{H}_S \to \mathcal{H}_F$ satisfying $T^*\mathcal{R}_F^{-1}T = -Q_O$. Such a factorization always exists—the "square-root" choice $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$ works for any Q_O—but physics-adapted choices yield better structural constants. This step fixes $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\mathrm{skew}} := \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$.

Step 3: Match the non-equilibrium perturbation. With $\mathcal{R}$ and $\mathcal{V}_{\mathrm{skew}}$ now determined by Step 2, the practitioner must choose $\mathcal{V}_{\mathrm{pert}}$ so that the full adiabatic-elimination formula $-P_S(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})\mathcal{R}^+(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})P_S$ reproduces $\mathcal{L}_O$. By (57), this amounts to the *consistency condition*

$$\mathcal{L}_O - Q_O\ =\ -\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}} + \mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S\ -\ \eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S, \tag{59}$$

which is an operator equation for $\mathcal{V}_{\mathrm{pert}}$, subject to the centering constraint $P_S\mathcal{V}_{\mathrm{pert}}P_S = 0$. In both examples below, the physical origin of the nonequilibrium driving provides a natural candidate for $\mathcal{V}_{\mathrm{pert}}$ that satisfies (59). In the Langevin example (Section 5.2), the non-conservative force F acts only on the momentum variable so that $\mathcal{V}_{\mathrm{pert}}P_S = 0$; accordingly, the cross terms reduce to a single contribution and the $\mathcal{O}(\eta^2)$ term vanishes. In the boundary-driven model (Section 5.3), all three terms in (57) are nonzero and their explicit forms appear in (74). The assembled lifted

generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies the centering condition (Definition 2.3) by construction, and Theorem 4.1 then provides an acceleration rate depending on the structural constants of the chosen factorization. A key feature of the reverse adiabatic elimination is that the lifted dynamics is *unbiased by construction*, as we now show.

Proposition 5.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (59)*. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the Schur complement (61) *satisfies $\gamma\, S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*
(iii) the lifted NESS satisfies $P_S h_\eta = g_\eta$ for all $\gamma > 0$ and $\eta \le \eta^*(\gamma)$, i.e., the bias $\delta(\gamma) \equiv 0$.

The condition $P_F\mathcal{V}_{\text{pert}}P_F = 0$ is a free design choice: it does not appear in the consistency condition (59) *(which constrains only $P_F\mathcal{V}_{\text{pert}}P_S$ and $P_S\mathcal{V}_{\text{pert}}P_F$), so it can always be enforced without affecting the effective dynamics.*

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F\mathcal{V}_{\text{skew}}P_F = 0$. Combined with $P_F\mathcal{V}_{\text{pert}}P_F = 0$, we get $P_F\mathcal{V}P_F = P_F(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_F = 0$. (ii) With $P_F\mathcal{V}P_F = 0$, the Schur complement (61) reduces to $S(\gamma) = -P_S\mathcal{V}(\gamma\mathcal{R}_F)^{-1}\mathcal{V}\,P_S = \gamma^{-1}\mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma)u = 0$ becomes $\mathcal{L}_O u = 0$, so $u = g_\eta$. □

Proposition 5.1 shows that the bias is not an inherent obstruction for designed lifts: by keeping the perturbation block-off-diagonal ($P_F\mathcal{V}_{\text{pert}}P_F = 0$), one eliminates the accuracy floor entirely. We stress that this condition does *not* trivialise the non-equilibrium structure: $\mathcal{L}_O = -P_S\mathcal{V}\,\mathcal{R}^+\mathcal{V}\,P_S$ depends only on the off-diagonal blocks $P_S\mathcal{V}_{\text{pert}}P_F$ and $P_F\mathcal{V}_{\text{pert}}P_S$, which remain unconstrained and keep $\mathcal{L}_O$ non-self-adjoint whenever $\eta \neq 0$. The boundary-driven example below satisfies this structure exactly, while the Langevin example exhibits no detectable bias at numerical resolution (Remark 5.6). The resulting algorithm is simple: run the lifted dynamics at γ_{opt} until the desired accuracy is reached.

Remark 5.2 (Kernel consistency and invariant states)*.* We verify that the lifted and effective dynamics share the same stationary state.

Equilibrium case ($\eta = 0$): exact identity for all $\gamma > 0$. Let $f \in \ker(\mathcal{L}|_{\eta=0}) = \ker(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}})$. Taking $\operatorname{Re}\langle f, \mathcal{L}f\rangle = \gamma\langle f, \mathcal{R}f\rangle = 0$ (using skew-adjointness of $\mathcal{V}_{\text{skew}}$) forces $f \in \ker(\mathcal{R}) = \mathcal{H}_S$. Then $\mathcal{V}_{\text{skew}}f = 0$, which entails $Tf = P_F\mathcal{V}_{\text{skew}}P_S f = 0$ and thus $Q_O f = -T^*\mathcal{R}_F^{-1}Tf = 0$. Therefore

$$\ker(\mathcal{L}|_{\eta=0}) = \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \forall\, \gamma > 0. \tag{60}$$

This is the analogue of [19, Lemma 2.10, eq. (2.17)].

Non-equilibrium case ($\eta \neq 0$). When $\eta \neq 0$ and $P_F\mathcal{V}P_F \neq 0$, the NESS $h_\eta \in \ker(\mathcal{L})$ acquires a fast component $P_F h_\eta = \mathcal{O}(\eta/\gamma)$, and the slow marginal $u = P_S h_\eta$ satisfies the Schur complement equation

$$S(\gamma)\, u = 0, \qquad S(\gamma) := -P_S\mathcal{V}\big(\gamma\mathcal{R}_F + P_F\mathcal{V}P_F\big)^{-1}\mathcal{V}\,P_S. \tag{61}$$

Expanding in a Neumann series:

$$\gamma\, S(\gamma) = \mathcal{L}_O + \gamma^{-1}\, P_S\mathcal{V}\,\mathcal{R}^+\,(P_F\mathcal{V}P_F)\,\mathcal{R}^+\,\mathcal{V}\,P_S + \mathcal{O}(\gamma^{-2}). \tag{62}$$

Thus $P_S h_\eta \to g_\eta$ as $\gamma \to \infty$ (the Zeno/adiabatic limit). In general the bias $\delta(\gamma) := \|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. However, when $P_F\mathcal{V}P_F = 0$ (as guaranteed by Proposition 5.1 for designed lifts), the expansion collapses to $\gamma\, S(\gamma) = \mathcal{L}_O$ and $\delta(\gamma) \equiv 0$.

Since the lifted dynamics converges to h_η, the approximation error for the overdamped NESS g_η satisfies the triangle inequality

$$\|P_S e^{t\mathcal{L}} f - g_\eta\|_{\mathcal{H}_S} \;\le\; \underbrace{e^{-\lambda_\eta t}\,\|f - h_\eta\|}_{\text{convergence error}} \;+\; \underbrace{\|P_S h_\eta - g_\eta\|_{\mathcal{H}_S}}_{\text{bias } \delta(\gamma)}. \tag{63}$$

For lifts satisfying the conditions of Proposition 5.1, $\delta(\gamma) = 0$ and the accelerated mixing time is

$$T_{\mathrm{mix}}(\varepsilon) \;=\; \mathcal{O}\Big(\frac{\log(1/\varepsilon)}{\sqrt{\lambda(Q_O)}}\Big), \qquad \forall\, \varepsilon > 0. \tag{64}$$

Algorithm 1 summarises the procedure.

Algorithm 1 Accelerated NESS Approximation via Lifting

Require: Target effective generator $\mathcal{L}_O$ on $\mathcal{H}_S$, target accuracy $\varepsilon > 0$
Ensure: Approximation of g_η (overdamped NESS) to accuracy ε

Step 1: Construct lifted dynamics (Section 5.1)
Factor $Q_O = -T^*\mathcal{R}_F^{-1}T$ and set $\mathcal{V}_{\mathrm{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$
Choose $\mathcal{V}_{\mathrm{pert}}$ satisfying (59) with $P_F\mathcal{V}_{\mathrm{pert}}P_F = 0$ ▷ Zero bias (Prop. 5.1)

Step 2: Compute optimal friction
$\gamma_{\mathrm{opt}} \leftarrow \sqrt{\lambda(Q_O)} \cdot \Theta^{-1}\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}} + \frac{1}{2}}$
$\lambda_\eta \leftarrow \frac{2}{3}\big(\Lambda_{\mathrm{hypo}}(\gamma_{\mathrm{opt}}) - \eta\, B_{\mathrm{pert}}(\gamma_{\mathrm{opt}})\big)$

Step 3: Run lifted dynamics (rate $\sim \sqrt{\lambda(Q_O)}$)
$T \leftarrow \log\big(\sqrt{3}\|f_0\|/\varepsilon\big)/\lambda_\eta$
return $P_S\big(e^{T\mathcal{L}}h_0\big)$ ▷ ε-approximation of g_η

Remark 5.3 (Two-phase extension for general models). For natural physical models where $P_F\mathcal{V}P_F \neq 0$ (and the conditions of Proposition 5.1 do not hold), a nonzero accuracy floor $\varepsilon_* := \delta(\gamma_{\mathrm{opt}}) = \mathcal{O}(\eta/\sqrt{\lambda(Q_O)})$ may arise. In this case, the error (63) saturates at ε_* and the lifted dynamics alone cannot reach accuracy $\varepsilon < \varepsilon_*$. A *two-phase strategy* resolves this: run the lifted dynamics at γ_{opt} for time $T_1 = \log(\|f_0\|/\varepsilon_*)/\lambda_\eta$, then switch to the unbiased overdamped generator $\mathcal{L}_O$ for time $T_2 = \log(\varepsilon_*/\varepsilon)/\lambda(\mathcal{L}_O)$. The total time $T_1 + T_2$ is at most $\log(\|f_0\|/\varepsilon)/\lambda(\mathcal{L}_O) = T_{\mathrm{OD}}(\varepsilon)$, with strict improvement whenever $\lambda_\eta > \lambda(\mathcal{L}_O)$. This extension is not needed for either of the examples in this paper (Remark 5.6).

Remark 5.4 (Connection to the literature). The reverse adiabatic elimination recipe mirrors, in continuous-time Markov generators, Szegedy's duplicated-state-space quantum walk [24] and the momentum-augmentation strategy of nonreversible MCMC acceleration [18, 10]. The abstract lifting framework of [19] formalizes similar ideas through Conditions A–D (self-adjoint/skew decomposition, strict kernel enlargement, centering, and compatibility of the overdamped generator with the reduced invariant state); our Assumption 3.1 plays an analogous role.

5.2. **Example 1: Classical Nonequilibrium Langevin Dynamics.** We specialize the abstract framework to a kinetic Fokker–Planck operator with non-gradient forcing on the torus $\mathbb{T}^d$.

Setup. Consider a particle on the phase space $\mathcal{X} = \mathbb{T}^d \times \mathbb{R}^d$ ($d \ge 1$), subject to a smooth conservative potential $U \in C^\infty(\mathbb{T}^d)$ and a non-conservative external force $F \in C^\infty(\mathbb{T}^d; \mathbb{R}^d)$. Let μ_0 be the Gibbs measure of the unperturbed dynamics:

$$d\mu_0(q,p) = Z^{-1}e^{-\beta(\frac{1}{2}|p|^2 + U(q))}\,dq\,dp, \tag{65}$$

with factorization $d\mu_0 = d\nu(q)\, d\nu_p(p)$, where $d\nu \propto e^{-\beta U} dq$ and $d\nu_p \propto e^{-\frac{\beta}{2}|p|^2} dp$. We work in $\mathcal{H} = L^2(\mu_0)$; for $\eta \neq 0$ the invariant measure $\mu_\eta \neq \mu_0$ and convergence to the NESS means $L^2(\mu_0)$-decay toward its density h_η with respect to μ_0. All operator identities are first verified on the core $\mathcal{C}_{\mathrm{Lan}} := C^\infty(\mathbb{T}^d) \otimes \mathcal{S}(\mathbb{R}^d)$ and extended by density.

Generator decomposition. With slow projection $P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\, d\nu_p(p)$, the generator decomposes as $\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\mathrm{pert}}$:

$$\mathcal{V}_{\mathrm{skew}} := p \cdot \nabla_q - \nabla U(q) \cdot \nabla_p, \qquad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\mathrm{pert}} f := F(q) \cdot \nabla_p f. \tag{66}$$

Effective slow generator. Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\, d\nu_p = 0$, one computes via Definition 2.4:

$$\mathcal{L}_O \phi = \beta^{-1}\Delta_q \phi - (\nabla U - \eta F) \cdot \nabla_q \phi, \qquad \phi = \phi(q), \tag{67}$$

and the reference operator is $Q_O := \mathcal{L}_O|_{\eta=0} = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q$, self-adjoint on $L^2(\nu)$ with Dirichlet form $\langle \phi, (-Q_O)\phi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q \phi\|^2_{L^2(\nu)}$.

Proposition 5.5 (Verification of Assumption 3.1)**.** *The decomposition* (66) *satisfies Assumption 3.1.1–4 in $L^2(\mu_0)$, with structural constants:*

(i) $\lambda_R = 1$, $M_R = 1$ *(Ornstein–Uhlenbeck spectral gap and conditioning);*
(ii) $\kappa_{\mathrm{ell}} = \sqrt{d(d+1)}\,\sqrt{\rho_{\max}/\rho_{\min}} \cdot (1 + \beta\|\nabla U\|_{L^\infty}\sqrt{C_{\mathrm{P}}(\nu)})$, *where* $\rho_{\max}/\rho_{\min} = e^{\beta(\max U - \min U)}$ *and* $C_{\mathrm{P}}(\nu)$ *is the Poincaré constant of* ν*;*
(iii) the perturbation constants (a_1, b_1, a_2, b_2) *depend only on* $(\beta, d, \|F\|_{L^\infty})$*, independently of* $\lambda(Q_O)$*.*

The proof (verification of skew-adjointness, centering, flux regularity via Fourier–Parseval analysis, and p-multiplier energy bounds) is given in Appendix E.

Scaling consequences. Since all structural constants are independent of $\lambda(Q_O)$, Corollary 4.2 applies with $\Theta = \frac{1}{2}$: the optimal friction is $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, the stability threshold satisfies $\eta \leq \eta^*(\gamma_{\mathrm{opt}}) \asymp C(\beta, d, \|F\|_{L^\infty}) \cdot \sqrt{\lambda(Q_O)}$, and the accelerated rate scales as $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$. This recovers the kinetic scaling of [11]: their sharp estimate $\|e^{t\mathcal{L}^\dagger} f - h_\eta\| \leq Ce^{-\bar{\lambda}_\delta \min(\gamma, \gamma^{-1})t}\|f - h_\eta\|$ exhibits the same underdamped ($\mathcal{O}(\gamma)$), overdamped ($\mathcal{O}(\gamma^{-1})$), and optimal ($\gamma \sim 1$ when $\lambda(Q_O) = \mathcal{O}(1)$) regime structure captured by $\Lambda_{\mathrm{hypo}}(\gamma)$ in Theorem 4.1. The new content here is the explicit stability threshold: the non-conservative force F (proportional to η) is admissible provided $\eta \leq \eta^* \asymp \sqrt{\lambda(Q_O)}$.

Numerical verification. We validate the predicted scaling by Galerkin spectral computations in the Fourier–Hermite basis [11]. To ensure $L^2(\mu_0)$-orthonormality, we use the Boltzmann-weighted Fourier modes of [11]:

$$G_k(q) = \sqrt{\frac{Z_\nu}{2\pi}}\, e^{ikq + \beta U(q)/2}, \qquad Z_\nu = \int_0^{2\pi} e^{-\beta U}\, dq,$$

so that $\{\psi_{nk}(q, p) := G_k(q)\, H_n(p)\}$ is an ONB in $L^2(\mu_0)$, where H_n are normalised Hermite functions in $L^2(\nu_p)$. For the rotor $U(q) = U_0(1 - \cos q)$, the generator matrix elements in this basis take the explicit form (with $m = 1$):

$$\langle \psi_{n'k'}, \mathcal{L}\, \psi_{nk}\rangle = \sqrt{\beta(n+1)}\, \delta_{n',n+1}\Big[\tfrac{ik}{\beta}\delta_{kk'} + \tfrac{u_{k-k'}}{2}\Big] + \sqrt{\beta n}\, \delta_{n',n-1}\Big[\big(\tfrac{ik}{\beta} + \eta F\big)\delta_{kk'} - \tfrac{u_{k-k'}}{2}\Big] - \gamma n\, \delta_{nn'}\delta_{kk'}, \tag{68}$$

where u_j are the Fourier coefficients of $U'(q)$ ($u_{\pm 1} = \mp iU_0/2$ for the rotor). The potential coupling enters *both* raising and lowering Hermite transitions with opposite signs, giving a balanced Galerkin truncation. We take $d = 1$, $\beta = 100$, $F = 0.05$, with $K = 8$ Fourier and $N = 16$ Hermite modes. The scan uses $U_0 \in \{0.01, 0.02, 0.05, 0.08, 0.1\}$ (so that $\beta U_0 \leq 10$), $\gamma \in [0.05, 3.0]$ (26 points), and $\eta \in \{0, 0.1, 1\}$.

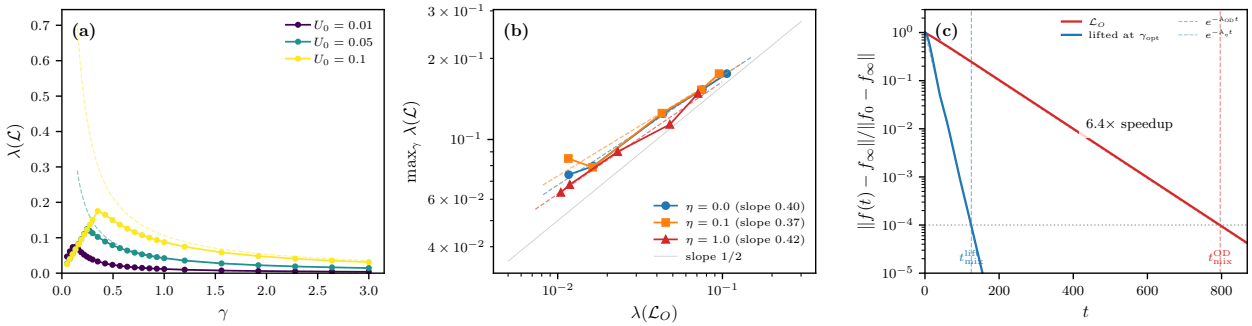

FIGURE 1. Numerical verification for Example 5.2. **(a)** $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for $U_0 \in \{0.01, 0.05, 0.1\}$; dashed curves show the overdamped ceiling $\lambda(\mathcal{L}_{\rm Smol})/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$ for $\eta \in \{0, 0.1, 1\}$; fitted slopes are shown in the legend. **(c)** Mixing-time comparison ($U_0 = 0.01$, $\eta = 0$): lifted dynamics at $\gamma_{\rm opt}$ (blue; no detectable bias at numerical resolution) vs. $\mathcal{L}_O$ (red). Dashed curves show the exponential references $e^{-\lambda t}$ at the respective spectral gaps. Vertical lines mark the ε-mixing times at $\varepsilon = 10^{-4}$ (dotted horizontal); both curves converge to the same invariant state (verified at $\eta = 0.1$: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$). Parameters: $\beta = 100$, $F = 0.05$, $K = 8$, $N = 16$.

As a discretization check, the free case $U_0 = 0$ recovers the exact eigenvalues $\lambda_{n,k} = n\gamma + k^2/(\beta\gamma)$ [15] to relative error $\lesssim 10^{-15}$. We also cross-validate against an unweighted Fourier–Hermite Galerkin basis (ONB in $L^2(dq\, d\nu_p)$ rather than $L^2(\mu_0)$): the two formulations agree to machine precision for $\beta U_0 \le 1$ and to 0.01% for $\beta U_0 = 5$; at $\beta U_0 = 10$ the unweighted basis produces spurious positive eigenvalues, confirming the necessity of the Boltzmann-weighted modes (68).

Figure 1(a) confirms the three-regime structure: underdamped branch $\lambda(\mathcal{L}) \propto \gamma$ at small friction, overdamped ceiling $\lambda(\mathcal{L}) \approx \lambda(Q_O)/\gamma$ at large friction, and a crossover peak at $\gamma_{\rm opt}$ that increases with U_0. Panel (b) shows the log–log scaling $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$; the fitted slope at $\eta = 0$ is consistent with the quadratic speedup $\max_\gamma \lambda(\mathcal{L}) \propto \sqrt{\lambda(\mathcal{L}_O)}$. At nonzero η, $\lambda(\mathcal{L}_O)$ shifts slightly from $\lambda(Q_O)$ due to the non-conservative force. For $\eta = 0.1$ all sampled friction values are stable, whereas for $\eta = 1$ instability appears only in the very low-friction regime; the intermediate/large-friction branch remains stable and follows the same scaling trend. Panel (c) compares the ε-mixing times at $\varepsilon = 10^{-4}$ for $U_0 = 0.01$, $\eta = 0$: the lifted dynamics at $\gamma_{\rm opt}$ (Algorithm 1) achieves $t_{\rm mix}^{\rm lift} \approx 125$, versus $t_{\rm mix}^{\rm OD} \approx 796$ for pure $\mathcal{L}_O$, giving a speedup of $\approx 6.4\times$. In this computation no accuracy floor is observed at the tested resolution, and the speedup persists down to $\varepsilon = 10^{-4}$. We verify numerically at $\eta = 0.1$ that both dynamics converge to the same invariant state: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$.

5.3. **Example 2: A Finite-Dimensional Boundary-Driven Toy Model.** We construct a two-qubit model in the spirit of boundary-driven constructions [22, 3, 17], illustrating that the framework handles non-self-adjoint effective slow generators. Since $\dim \mathcal{H} = 16$, all operators are bounded and domain questions are vacuous.

Setup. The fast ancilla and slow system are both qubits ($\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$). We work on $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$ with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The generator decomposes as $\mathcal{L} = \mathcal{V}_{\rm skew} + \gamma\mathcal{R} + \eta\mathcal{V}_{\rm pert}$. The *fast dissipator* is $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$, where $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$ with $\Gamma > 0$; since σ_j are unitary, $\mathcal{R}_F(\sigma_j) = -4\Gamma\sigma_j$, giving $\lambda_R = 4\Gamma$. The *Hamiltonian transport* is $\mathcal{V}_{\rm skew} X = -i[g\sigma_z \otimes \sigma_z, X]$ for

$g \in \mathbb{R}$, and the *perturbation* is $\mathcal{V}_{\text{pert}} = \sigma_z \otimes \mathcal{G}_S$ with $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$ (amplitude-damping), where $\sigma_- = \frac{1}{2}(\sigma_x - i\sigma_y)$. Since $\operatorname{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$, so the centering condition holds.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space).

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = \sigma_z \otimes \Big(- ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big) \in \mathcal{H}_F. \tag{69}$$

Step 2 (Poisson equation on the fast ancilla). In finite dimensions, the Moore–Penrose pseudoinverse can be written as

$$\mathcal{R}^+ := -\int_0^\infty \left(e^{s\mathcal{R}} - P_S\right) ds, \tag{70}$$

so that $\mathcal{R}\mathcal{R}^+ = \mathrm{id} - P_S = P_F$ and $P_S\mathcal{R}^+ = \mathcal{R}^+P_S = 0$. In particular, since $\mathcal{R}_F(\sigma_z) = -\kappa\,\sigma_z$ with $\kappa := 4\Gamma$, we have $\mathcal{R}^+(\sigma_z \otimes Y) = \mathcal{R}_F^+(\sigma_z) \otimes Y = -(1/\kappa)\sigma_z \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa}\sigma_z \otimes \Big(- ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big). \tag{71}$$

Step 3 (return map and projection). Since $\mathcal{V}_{\text{skew}}(\sigma_z \otimes Y) = -ig\,\mathbf{1}_F \otimes [\sigma_z, Y]$ and $\mathcal{V}_{\text{pert}}(\sigma_z \otimes Y) = \mathbf{1}_F \otimes \mathcal{G}_S(Y)$, we compute

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\,\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X \tag{72}$$

$$= \frac{1}{\kappa}\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z, [\sigma_z, X_S]] + i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] + i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) - \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{73}$$

Projecting with $P_S(\mathbf{1}_F \otimes Y) = \mathbf{1}_F \otimes Y$ yields

$$\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z, [\sigma_z, X_S]] \tag{74}$$

$$+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \tag{75}$$

$$+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \tag{76}$$

$$- \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{77}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z, [\sigma_z, X_S]]. \tag{78}$$

Non-self-adjointness on the slow space. Since $\pi_S \propto \mathbf{1}_S$, the $L^2(\pi_S)$ inner product is proportional to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}^\dagger[\sigma_+] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $L^2(\pi_S)$. Consequently, the $\mathcal{O}(\eta)$ mixed operator

$$\mathcal{B} := ig\,\big(\operatorname{ad}_{\sigma_z} \circ \mathcal{G}_S + \mathcal{G}_S \circ \operatorname{ad}_{\sigma_z}\big), \qquad \operatorname{ad}_{\sigma_z}(X) := [\sigma_z, X], \tag{79}$$

is generically not self-adjoint either (unless additional commutation/self-adjointness relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$.

Verification of Assumption 3.1. All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 3.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 3.1.2 holds since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.

(3) Assumption 3.1.3 holds with $M_R := \|\mathcal{R}\|$. Moreover, in this toy model the flux regularity bound (11) can be checked explicitly. For $\psi = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$ one has $g_\psi = P_F \mathcal{V}_{\text{skew}} P_S \psi = -ig\,\sigma_z \otimes [\sigma_z, X_S]$ and $(-\mathcal{R}^+)g_\psi = (1/\kappa)g_\psi$. Therefore

$$P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi = 0, \tag{80}$$

$$(-Q_O)\psi = P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi = \frac{g^2}{\kappa}\, \mathbf{1}_F \otimes [\sigma_z, [\sigma_z, X_S]]. \tag{81}$$

Since the left-hand side is identically zero, (11) holds (e.g. with $\kappa_{\text{ell}} := 1$).

(4) For Assumption 3.1.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger)$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (14)–(15) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

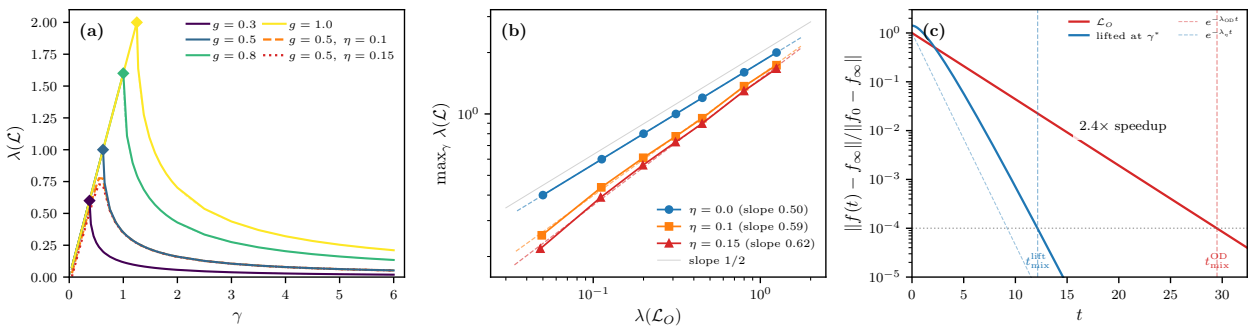

FIGURE 2. Numerical verification of Example 5.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ: solid curves at $\eta = 0$ for $g \in \{0.3, 0.5, 0.8, 1.0\}$, dashed curves the exact analytic formula (82), diamond markers the exact peaks $2g$ at $\gamma^* = g/\Gamma$; dashed/dotted coloured curves show the stable gap at $\eta = 0.10$ and $\eta = 0.15$ for $g = 0.5$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$ for $\eta \in \{0, 0.10, 0.15\}$; fitted slopes are shown in the legend. **(c)** Mixing-time comparison ($g = 0.5$, $\eta = 0$): lifted dynamics at γ^* (blue; $\varepsilon_* = 0$ by Proposition 5.1, confirming zero bias) vs. $\mathcal{L}_O$ (red). Dashed curves show exponential references $e^{-\lambda t}$. Vertical lines mark the ε-mixing times at $\varepsilon = 10^{-4}$ (dotted horizontal); both curves converge to the same invariant state (verified at $\eta = 0.1$: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-15}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.2, \ldots, 1.0\}$.

Numerical verification. We assemble the exact 16×16 Liouville-space generator and compute its spectral gap. The scan uses $g \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0\}$, $\Gamma = 0.8$ ($\kappa = 4\Gamma = 3.2$), $\gamma \in [0.05, 6.0]$ (38 points, including the exact peaks $\gamma^* = g/\Gamma$), and $\eta \in \{0, 0.10, 0.15\}$. The $\eta = 0$ case admits an exact analytic solution: the slow-sector eigenvalues of $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}}$ are given by the 2×2 block

$$\lambda_\pm = \tfrac{1}{2}\big(-\gamma\kappa \pm \sqrt{(\gamma\kappa)^2 - 16g^2}\big). \tag{82}$$

For $\gamma < \gamma^* := g/\Gamma$, the eigenvalues are complex with gap $\gamma\kappa/2$ (underdamped); at $\gamma = \gamma^*$ they coalesce at the peak gap $2g$; for $\gamma > \gamma^*$, the gap is the slow real eigenvalue $(\gamma\kappa - \sqrt{(\gamma\kappa)^2 - 16g^2})/2 \approx g^2/(\Gamma\gamma)$ (overdamped). The numerical eigenvalues agree with (82) to machine precision ($|\Delta| \lesssim 10^{-8}$), as confirmed by the sanity check in Figure 2(a). The exact maximizer $\gamma^* = g/\Gamma$ differs from $\sqrt{\lambda(\mathcal{L}_O)} = g/\sqrt{\Gamma}$ by a model-dependent constant factor $\Gamma^{-1/2}$; this is consistent with the theorem-level prediction $\gamma_{\text{opt}} \asymp \sqrt{\lambda(\mathcal{L}_O)}$.

Panel (b) displays $\max_\gamma \lambda(\mathcal{L}) = 2g$ versus $\lambda(\mathcal{L}_O) = g^2/\Gamma$, yielding slope exactly 1/2 in the log–log plane (since $\log(2g) = \frac{1}{2}\log(g^2/\Gamma) + \text{const}$), confirming $\max_\gamma \lambda(\mathcal{L}) \propto \sqrt{\lambda(\mathcal{L}_O)}$. (At $\eta = 0$ the effective generator reduces to $\mathcal{L}_O = Q_O$. For $\eta \in \{0.10, 0.15\}$, the stable slow gap remains very close to g^2/Γ in our computations.) For $\eta \neq 0$, the perturbation $\mathcal{V}_{\text{pert}}$ introduces an $\mathcal{O}(\eta^2/\kappa)$ instability in $\mathcal{L}_O$ (a single eigenvalue with positive real part), which we exclude when defining the stable spectral gap $\lambda_{\text{stab}}(\mathcal{L})$. The $\eta \neq 0$ curves in panel (a) and data in panel (b) use this stable gap, showing that the dominant convergence physics is essentially unaffected by the perturbation: the fitted slope remains close to 1/2. Panel (c) compares the ε-mixing times at $\varepsilon = 10^{-4}$ for $g = 0.5$, $\eta = 0$: the lifted dynamics at γ^* (Algorithm 1) achieves $t_{\text{mix}}^{\text{lift}} \approx 12.2$ versus $t_{\text{mix}}^{\text{OD}} \approx 29.5$ for pure $\mathcal{L}_O$, a speedup of $\approx 2.4\times$. The mixing-time speedup ($2.4\times$) is smaller than the spectral-gap ratio ($3.2\times$) because the lifted semigroup incurs a transient prefactor from non-normal coupling to the fast modes; by contrast, the self-adjoint overdamped semigroup decays as a pure exponential. Convergence to the same invariant state is verified numerically at $\eta = 0.1$: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \sim 10^{-15}$.

Remark 5.6 (Bias behaviour in the two examples). The two examples exhibit very small lifted-to-overdamped bias, but with different levels of rigor. In the boundary-driven model, the sole fast-slow coupling channel is the σ_z mode, and one checks that $P_F \mathcal{V} P_F = 0$ on this mode (since both $\mathcal{V}_{\text{skew}}$ and $\mathcal{V}_{\text{pert}}$ map $\sigma_z \otimes Y$ back to $\mathcal{H}_S$). This makes the Schur complement (61) γ-independent: $\gamma S(\gamma) = \mathcal{L}_O$ exactly, so $P_S h_\eta = g_\eta$ for all $\gamma > 0$. In the Langevin model, $\mathcal{V}_{\text{pert}} P_S = 0$ ($F \cdot \partial_p$ kills the ($n = 0$)-Hermite ground state), and the return path constraints suppress leading bias contributions; in the Galerkin computation used here, we observe $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$ at $\eta = 0.1$. Consequently, Algorithm 1 shows no detectable accuracy floor in both numerical experiments; an exact statement $\varepsilon_* = 0$ is proved for the boundary-driven model, while for Langevin the evidence here is numerical. For general models where $P_F \mathcal{V} P_F \neq 0$, a two-phase strategy is available (Remark 5.3).

6. Conclusion and open problems

We have developed a quantitative hypocoercive framework for generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on a slow–fast Hilbert decomposition, providing an explicit characterisation of the interplay between the friction γ, the reversible backbone gap $\lambda(Q_O)$, and the non-equilibrium driving strength η. First, Theorem 4.1 provides an *explicit η-dependent rate*: $\lambda_\eta \geq \frac{2}{3}(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma))$, degrading linearly in η with fully computable slope. Second, Corollary 4.2 establishes a *stability threshold* that scales as $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ at optimal friction—systems with faster reversible mixing tolerate stronger non-equilibrium driving. Third, Theorem 4.4 establishes an *overdamped ceiling*: the rate degenerates to $\mathcal{O}(\gamma^{-1}\lambda(\mathcal{L}_O)) + \mathcal{O}(\eta)$ for large γ, confirming that the acceleration is inherently a finite-friction phenomenon. Within the stability regime, the full effective generator satisfies $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$ (Remark 3.4), so the analysis is self-consistent whether one measures convergence relative to the reversible backbone Q_O or the full effective generator $\mathcal{L}_O$. For lifts constructed via the reverse adiabatic elimination recipe (Section 5.1), Proposition 5.1 shows that the bias can always be eliminated by design ($P_F \mathcal{V}_{\text{pert}} P_F = 0$), giving $\varepsilon_* = 0$ and the full quadratic speedup for all target accuracies $\varepsilon > 0$. The boundary-driven example realizes this exactly, while the Langevin example exhibits the same behaviour numerically up to machine precision (Remark 5.6). For general natural models where $P_F \mathcal{V} P_F \neq 0$, a two-phase extension is available (Remark 5.3).

We conclude with several directions for future investigation.

(1) **Sharp stability thresholds.** The threshold η^* in Theorem 4.1 is derived from a Lyapunov argument and involves structural constants that may not be sharp. Numerically, the accelerated regime appears to persist well beyond η^*; sharpening this threshold is an important open problem. In particular, for models where the non-reversible perturbation *enhances* the effective gap (e.g. optimally rotated diffusions [10]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a

refined analysis incorporating this enhancement could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{83}$$

(2) Optimal factorization for the reverse adiabatic elimination. The reverse adiabatic elimination recipe (Section 5.1) shows that any target Q_O can be lifted, but the resulting structural constants—and hence the stability threshold—depend on the choice of auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the factorization that maximises η^* (and minimises K_{geom}, κ_{ell}) in infinite-dimensional settings where domain issues (Remark 3.2) constrain the couplings remains open. In classical sampling, multilevel ("replica") methods and cascaded Langevin lifts [20] have been shown to provide iterated acceleration. Extending the present framework—including the non-equilibrium stability analysis—to a chain of nested slow–fast decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$ could yield explicit multilevel stability thresholds and rate bounds.

References

1. Giovanni Brigati, Francis Lörler, and Lihan Wang, *Hypocoercivity meets lifts*, arXiv preprint arXiv:2412.10890, 2025.
2. Yu Cao, Jianfeng Lu, and Lihan Wang, *On explicit l^2-convergence rate estimate for underdamped langevin dynamics*, Archive for Rational Mechanics and Analysis **247** (2023), no. 5.
3. Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Boundary-driven quantum systems near the zeno limit: steady states and long-time behavior*, arXiv preprint arXiv:2512.12825, 2025.
4. Sourav Chatterjee, *Spectral gap of nonreversible Markov chains*, arXiv preprint arXiv:2310.10876, 2025.
5. Fang Chen, László Lovász, and Igor Pak, *Lifting Markov chains to speed up mixing*, Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing (New York), STOC '99, ACM, 1999, pp. 275–281.
6. Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for linear kinetic equations conserving mass*, Transactions of the American Mathematical Society **367** (2015), no. 6, 3807–3828.
7. Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for kinetic equations with linear relaxation terms*, 2009.
8. Andreas Eberle, Arnaud Guillin, Leo Hahn, Francis Lörler, and Manon Michel, *Convergence of non-reversible Markov processes via lifting and flow Poincaré inequality*, arXiv preprint arXiv:2503.04238, 2025.
9. Andreas Eberle and Francis Lörler, *Non-reversible lifts of reversible diffusion processes and relaxation times*, arXiv preprint arXiv:2402.05041, 2024.
10. Chii-Ruey Hwang, Sheng-Yao Hwang-Ma, and Shuenn-Jyi Sheu, *Accelerating diffusions*, The Annals of Applied Probability **15** (2005), no. 2, 1433–1444.
11. Alessandra Iacobucci, Stefano Olla, and Gabriel Stoltz, *Convergence rates for nonequilibrium Langevin dynamics*, arXiv preprint arXiv:1702.03685, 2017.
12. Marcelo Janovitch, Matteo Brunelli, and Patrick P. Potts, *Active quantum reservoir engineering: Using a qubit to manipulate its environment*, arXiv preprint arXiv:2505.16898, 2025.
13. Tosio Kato, *Perturbation theory for linear operators*, reprint of the 1980 edition ed., Classics in Mathematics, Springer-Verlag, Berlin, 1995.
14. E. M. Kessler, *Generalized Schrieffer-Wolff formalism for dissipative systems*, Physical Review A **86** (2012), no. 1.
15. Sergeı M Kozlov, *Effective diffusion in the fokker-planck equation*, Mathematical notes of the Academy of Sciences of the USSR **45** (1989), no. 5, 360–368.
16. Gabriel T. Landi, Dario Poletti, and Gernot Schaller, *Nonequilibrium boundary-driven quantum systems: Models, methods, and properties*, Rev. Mod. Phys. **94** (2022), 045006.
17. François-Marie Le Régent and Pierre Rouchon, *Adiabatic elimination for composite open quantum systems: reduced model formulation and numerical simulations*, arXiv preprint arXiv:2303.05089, 2023.
18. Tony Lelièvre, Francis Nier, and Grigorios A. Pavliotis, *Optimal non-reversible linear drift for the convergence to equilibrium of a diffusion*, Journal of Statistical Physics **152** (2013), no. 2, 237–274.
19. Bowen Li and Jianfeng Lu, *Speeding up quantum Markov processes through lifting*, arXiv preprint arXiv:2505.12187, 2025.
20. Pierre Monmarché and Mouad Ramil, *Overdamped limit at stationarity for non-equilibrium Langevin diffusions*, arXiv preprint arXiv:2110.01238, 2022.
21. Daniel Paulin, *Concentration inequalities for markov chains by marton couplings and spectral methods*, arXiv preprint arXiv:1212.2015, 2018, Introduces the pseudo spectral gap and relates it to multiplicative reversiblization for non-reversible Markov chains.

22. Vladislav Popkov, Simon Essink, Carlo Presilla, and Gunter Schütz, *Effective quantum Zeno dynamics in dissipative quantum systems*, Physical Review A **98** (2018), no. 5.
23. Vladislav Popkov, Carlo Presilla, and Mario Salerno, *Manifolds of exceptional points and effective Zeno limit of an open two-qubit system*, arXiv preprint arXiv:2412.13674, 2025.
24. Mario Szegedy, *Quantum speed-up of Markov chain based algorithms*, 45th Annual IEEE Symposium on Foundations of Computer Science, IEEE, 2004, pp. 32–41.
25. Masaaki Tokieda and Angela Riva, *Time-convolutionless master equation applied to adiabatic elimination*, Physical Review A **111** (2025), no. 5.
26. Cédric Villani, *Hypocoercivity*, Mem. Amer. Math. Soc., vol. 202, American Mathematical Society, Providence, RI, 2009. MR 2562709

Appendix A. Proofs for Preliminaries

This appendix contains the proofs of the spectral bounds and hypocoercivity characterizations stated in Section 2. The arguments are carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$. In the noncommutative realization, this corresponds to $\mathcal{H} = L^2(\sigma)$ and $\mathcal{H}_0 = \{X : \mathrm{tr}(\sigma X) = 0\}$.

A.1. **Proof of Lemma 2.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part* $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$*, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{84}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{85}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy-Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma 2.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

(⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \leq -\lambda_S \|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{86}$$

For any $X_t = \mathcal{P}_t X_0 \in \mathcal{H}_0$, the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \leq -\lambda_S \|X_t\|^2. \tag{87}$$

Integrating this differential inequality yields $\|X_t\| \leq e^{-\lambda_S t}\|X_0\|$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

(⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \leq e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. The Hille-Yosida theorem (or simple differentiation at $t = 0$) implies that the generator satisfies the dissipativity condition:

$$\Re\langle X, \mathcal{L}X\rangle \leq -\nu\|X\|^2, \quad \forall X \in \mathcal{H}_0. \tag{88}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \leq -\nu\|X\|^2. \tag{89}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \leq -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B. Proof of Lemma 3.8

This appendix establishes the structural bounds of Lemma 3.8.

Proof. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. By Assumption 3.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{90}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (9), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\text{skew}}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\mu f \in \mathcal{H}_S$. The defining equation is $(\mu - Q_O)u = P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\mu\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{91}$$

Let $v := \mathcal{V}_{\text{skew}}u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (90):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{92}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\mu\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{93}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\mu\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{94}$$

so $\mu\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\mu^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\mu^{-1/2}\,\|f\|. \tag{95}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\mathrm{dom}(Q_O) = \mathcal{H}_S$, so $u \in \mathrm{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \mathrm{dom}(Q_O)$; we may therefore apply the flux conditioning bound (9) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{96}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\mu\| \le k_{\mathcal{A}}\mu^{-1/2}$ and $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \le k_{\text{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\text{reg}} := \sqrt{M_R/\lambda_R}. \tag{97}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\mu\mathcal{V}_{\text{skew}}P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\mu^\dagger = \left((\mu - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\mu - Q_O)^{-1}. \tag{98}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F\mathcal{V}_{\text{skew}}^\dagger\mathcal{A}_\mu^\dagger = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\mu - Q_O)^{-1}. \tag{99}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\mu - Q_O)^{-1}g$. Define the macroscopic flux

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{100}$$

Then $\mathcal{T}^\dagger g = P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi$. Invoking the **Flux Regularity** condition from Assumption 3.1.3:

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\| \le \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{101}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}} \|(-Q_O)(\mu - Q_O)^{-1} g\|. \tag{102}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\mu - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\text{ell}} \|g\|. \tag{103}$$

This proves $\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| \le \kappa_{\text{ell}}$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption 3.1.4. We also recall the centered mapping property (31), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 3.1.4(a)):

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{104}$$

For the symmetric part, we use the energy bound (Assumption 3.1.4(b)):

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{105}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\mu f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\| \tag{106}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{107}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\mu$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{108}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \mu^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f\| \le (b_1 + b_2) k_{\mathcal{A}} \mu^{-1/2} \|f\|. \tag{109}$$

This establishes (28a) with $k_P(\mu) := (b_1 + b_2) k_{\mathcal{A}} \mu^{-1/2}$.

Forward estimate ($\mathcal{A}_\mu \mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\mu \mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^\dagger\| = \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\mu^\dagger g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\|\mathcal{V}_{\text{pert}}^\dagger w\| \le \|\mathcal{K}_{\text{pert}} w\| + \|\mathcal{S}_{\text{pert}} w\|$$

$$\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \tag{110}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\mu^\dagger g\| \le \|\mathcal{A}_\mu\| \|g\| \le k_{\mathcal{A}} \mu^{-1/2}. \tag{111}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2}w\|$):** Using $w = \mathcal{A}_\mu^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2}w\| = \|(-\mathcal{R})^{1/2}\mathcal{A}_\mu^\dagger g\| = \|(-\mathcal{R})^{-1/2}P_F\mathcal{V}_{\text{skew}}P_S(\mu - Q_O)^{-1}g\|. \tag{112}$$

Setting $\psi := (\mu - Q_O)^{-1}g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi, \tag{113}$$

$$\|(-\mathcal{R})^{-1/2}g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = -\langle \psi, Q_O\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2, \tag{114}$$

hence

$$\|(-\mathcal{R})^{1/2}w\| = \|(-Q_O)^{1/2}(\mu - Q_O)^{-1}g\| \le \frac{1}{2}\,\mu^{-1/2}\,\|g\|. \tag{115}$$

(The last step follows from the scalar bound $\sup_{x\ge 0}\frac{\sqrt{x}}{\mu+x} = \frac{1}{2\sqrt{\mu}}$.)

Substituting these estimates into (110) yields

$$\|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\mu^\dagger g\| \le \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\mu^{-1/2}\,\|g\|. \tag{116}$$

Since g is arbitrary, this establishes (28b) with $\tilde{k}_P(\mu) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\mu^{-1/2}$.

□

Appendix C. Proof of Theorem 4.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\mu > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle \mathcal{A}_\mu f, f\rangle, \tag{117}$$

where $\mathcal{A}_\mu = (\mu - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

C.1. **Preliminaries.** We strictly utilize the structural estimates established in Lemma 3.8. We recall the constants derived therein, depending solely on the parameters from Assumption 3.1 $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$. For convenience in the Lyapunov dissipation bounds below, we introduce the effective symmetric-perturbation coefficient

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{118}$$

which absorbs both the slow-mode contribution $b_2 + \frac{a_2}{2}$ and the fast-mode λ_R-amplification $\frac{a_2\lambda_R}{2}$ that arises below when applying $\|(-\mathcal{R})^{1/2}f\|^2 \ge \lambda_R\|f_F\|^2$ to the symmetric perturbation energy. The relevant bounds from Lemma 3.8 are the Solver Bound $\|\mathcal{A}_\mu\| \le k_{\mathcal{A}}\mu^{-1/2}$ with $k_{\mathcal{A}} := (2\sqrt{\lambda_R})^{-1}$, the Regularity estimates $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \le k_{\text{reg}} := \sqrt{M_R/\lambda_R}$ and $\|\mathcal{A}_\mu\mathcal{V}_{\text{skew}}P_F\| \le \kappa_{\text{ell}}$, and the Perturbation Control bounds

$$\|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\mu\| \le k_P(\mu) = (b_1+b_2)k_{\mathcal{A}}\,\mu^{-1/2}, \tag{119a}$$

$$\|\mathcal{A}_\mu\mathcal{V}_{\text{pert}}\| \le \tilde{k}_P(\mu) = \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}. \tag{119b}$$

We denote the spectral gap of the self-adjoint operator $-Q_O$ by $\lambda(Q_O)$.

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}} f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L} f_t$. In unbounded settings, the computations below are performed on a common invariant core $\mathcal{C} \subset \mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}(\mathcal{A}_\mu)$ and extended to the energy domain by density.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \mathrm{dom}(\mathcal{R})$,*

$$\Re\langle \mathcal{R} f, f\rangle = \langle \mathcal{R} f, f\rangle = -\|(-\mathcal{R})^{1/2} f\|^2. \tag{120}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle \mathcal{R} f, f\rangle = \Re\langle \mathcal{R} f_F, f_F\rangle. \tag{121}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle \mathcal{R} f, f\rangle = -\langle (-\mathcal{R}) f, f\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Moreover $\mathcal{R} f_S = 0$ and $\mathcal{R} f = \mathcal{R} f_F$. □

Let f_t satisfy $\partial_t f_t = \mathcal{L} f_t$. By differentiating (33) along the flow and using that $\mathcal{A}_\mu$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_t, \partial_t f_t\rangle \tag{122}$$

$$= \Re\langle \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_t, \mathcal{L} f_t\rangle. \tag{123}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{124}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L} f, f\rangle, \tag{125}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L} f\rangle. \tag{126}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R} f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{127}$$

Using Assumption 3.1.1 ($\mathcal{R}$ coercivity) and Assumption 3.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R} f, f\rangle \leq -\gamma\lambda_R \|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (15) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \leq \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \leq \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{128}$$

$$\leq \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{129}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R} f, f\rangle = \Re\langle \mathcal{R} f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Hence

$$\gamma\Re\langle \mathcal{R} f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \leq -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{130}$$

$$\leq -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{131}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \geq \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{132}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L} f\rangle. \tag{133}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{134}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle, \tag{135}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\mathrm{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\mathrm{skew}} f\rangle, \tag{136}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\mathrm{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\mathrm{pert}} f\rangle. \tag{137}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle. \tag{138}$$

Since $\mathcal{R} f = \mathcal{R} f_F$ and $\mathcal{A}_\mu f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\mu f, f\rangle = 0. \tag{139}$$

Moreover $\mathcal{A}_\mu \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f_F, f_S\rangle. \tag{140}$$

Using $\mathcal{A}_\mu = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\mu \mathcal{R} f_F = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)\mathcal{R} f_F = -(\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}} f_F. \tag{141}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\,\big|\Re\langle (\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}} f_F, f_S\rangle\big| = \alpha\gamma\,\big|\Re\langle f_F, \mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S\rangle\big| \tag{142}$$

$$\le \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S\|. \tag{143}$$

To bound $\|\mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S\|$, set $\psi := (\mu - Q_O)^{-1} f_S \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\mathrm{skew}} P_S \psi \in \mathcal{H}_F. \tag{144}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\mathrm{skew}} \psi = 0$, hence $g_\psi = \mathcal{V}_{\mathrm{skew}}\psi = \mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S$. Applying the flux conditioning bound (9) yields

$$\|\mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \le M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{145}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\mathrm{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{146}$$

Therefore

$$\|\mathcal{V}_{\mathrm{skew}}(\mu - Q_O)^{-1} f_S\| \le \sqrt{M_R}\,\|(-Q_O)^{1/2}(\mu - Q_O)^{-1} f_S\|. \tag{147}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \ge 0$,

$$\|(-Q_O)^{1/2}(\mu - Q_O)^{-1}\| = \sup_{x\ge 0} \frac{\sqrt{x}}{\mu + x} = \frac{1}{2\sqrt{\mu}}, \tag{148}$$

which implies

$$\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\, \mu^{-1/2}\, \|f_S\|. \tag{149}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\, \frac{\sqrt{M_R}}{2}\, \mu^{-1/2}\, \|f_F\|\, \|f_S\|. \tag{150}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\, \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle + \alpha\, \Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\text{skew}} f\rangle, \tag{151}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\mu$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\mu = \mathcal{A}_\mu P_F$ (since P_F is on the right in (23)). Hence $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle \tag{152}$$

$$+ \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_F\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_F\rangle. \tag{153}$$

The last term vanishes because $\mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle. \tag{154}$$

In the *second* bracket we use $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$ and expand $\mathcal{V}_{\text{skew}} f$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\text{skew}} f\rangle = \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle + \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle. \tag{155}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\, \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \alpha\, \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle \tag{156}$$

$$+ \alpha\, \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle + \alpha\, \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle. \tag{157}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{skew}} P_S = 0$, we have $P_F \mathcal{V}_{\text{skew}} f_S = \mathcal{V}_{\text{skew}} f_S$. By the definition of $\mathcal{A}_\mu$ and Q_O this yields

$$\alpha \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle = \alpha\langle (\mu - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\, \langle (\mu - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{158}$$

$$\leq -\alpha\, \frac{\lambda(Q_O)}{\mu + \lambda(Q_O)}\, \|f_S\|^2. \tag{159}$$

Cross Terms. Using the Regularity estimates:

$$\alpha \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle + \alpha \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle \tag{160}$$

$$\leq \alpha(\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| + \|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|)\|f_F\|\|f_S\| \tag{161}$$

$$\leq \alpha(\kappa_{\text{ell}} + k_{\text{reg}})\|f_F\|\|f_S\|. \tag{162}$$

Fast–fast remainder. Using $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ and Cauchy–Schwarz,

$$\big|\alpha \Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle\big| = \alpha \big|\Re\langle -\mathcal{V}_{\text{skew}} \mathcal{A}_\mu f_F, f_F\rangle\big| \leq \alpha\, \|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|\, \|f_F\|^2 \leq \alpha\, k_{\text{reg}}\, \|f_F\|^2. \tag{163}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta \Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle + \alpha\eta \Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle. \tag{164}$$

We estimate each term using Lemma 3.8 and the centered mapping property (31).
Reverse term. Since $\mathcal{A}_\mu f \in \mathcal{H}_S$ and $\mathcal{V}_{\text{pert}}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle = \langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f_F\rangle$. Therefore,

$$\big|\alpha\eta \Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle\big| \leq \alpha\eta\, \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f\|\, \|f_F\| \leq \alpha\eta\, k_P(\mu)\, \|f\|\, \|f_F\| \tag{165}$$

$$\leq \alpha\eta\, k_P(\mu)\, \big(\|f_F\|^2 + \|f_F\|\, \|f_S\|\big). \tag{166}$$

Forward term. Since $\mathcal{A}_\mu \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\mu \mathcal{V}_{\rm pert} f, f\rangle = \langle \mathcal{A}_\mu \mathcal{V}_{\rm pert} f, f_S\rangle$. Therefore,

$$\left|\alpha\eta \Re\langle \mathcal{A}_\mu \mathcal{V}_{\rm pert} f, f\rangle\right| \le \alpha\eta \, \|\mathcal{A}_\mu \mathcal{V}_{\rm pert} f\| \, \|f_S\| \le \alpha\eta \, \tilde{k}_P(\mu) \, \|f\| \, \|f_S\| \tag{167}$$

$$\le \alpha\eta \, \tilde{k}_P(\mu) \left(\|f_S\|^2 + \|f_F\| \, \|f_S\|\right). \tag{168}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle X f_F, f_S\rangle| \le \|X\| \, \|f_F\| \, \|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt} \mathcal{E}_{\alpha,\mu}(f_t) \le -\mathbf{v}(t)^T \mathbf{M} \, \mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{169}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, μ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of μ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\rm eff}\big) - \alpha k_{\rm reg} - \alpha\eta k_P(\mu) \\ &= \gamma\lambda_R - \eta b_{2,\rm eff} - \alpha k_{\rm reg} - \alpha\eta (b_1 + b_2) k_{\mathcal{A}} \, \mu^{-1/2}. \end{aligned} \tag{170}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha \frac{\lambda(Q_O)}{\mu + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta \, \tilde{k}_P(\mu) \\ &= \alpha \frac{\lambda(Q_O)}{\mu + \lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta \left[\Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2}\Big)\mu^{-1/2}\right]. \end{aligned} \tag{171}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha \left[\gamma \frac{\sqrt{M_R}}{2} \, \mu^{-1/2} + (k_{\rm reg} + \kappa_{\rm ell}) + \eta\big(k_P(\mu) + \tilde{k}_P(\mu)\big)\right] \\ &= -\frac{1}{2}\alpha \left[k_{\rm reg} + \kappa_{\rm ell} + \left(\gamma \frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2}\Big)\right) \mu^{-1/2}\right]. \end{aligned} \tag{172}$$

This explicit form allows for direct asymptotic analysis by choosing $\mu \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)}. \tag{173}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\mu + \lambda(Q_O))$, while the cross term $|M_{12}^0|$ contains $\gamma\Theta\mu^{-1/2}$. A leading-order balance (maximising $\mu(\mu + \lambda(Q_O))^{-2}$ in the large-γ rate) selects

$$\mu_* := \lambda(Q_O). \tag{174}$$

At this value, $\lambda(Q_O)/(\mu_* + \lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\rm geom}, Z(\gamma)$ from (35)–(36), the unperturbed matrix entries at $\mu = \mu_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\rm reg}, \tag{175}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{176}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\, Z(\gamma). \tag{177}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\rm reg} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{178}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}}, \tag{179}$$

in agreement with (37), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\rm reg}}. \tag{180}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\rm reg}) + \frac{1}{2}\alpha_*$. Dropping the non-positive term $-\alpha_* k_{\rm reg}$ gives the conservative upper bound

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \le \gamma\lambda_R + \tfrac{1}{2}\alpha_* = \gamma\lambda_R\left(1 + \frac{\frac{1}{2}}{Z(\gamma)^2 + 2k_{\rm reg}}\right). \tag{181}$$

Applying (173) yields

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \ge \frac{\frac{1}{4}\,\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg} + \frac{1}{2}} =: \Lambda_{\rm hypo}(\gamma), \tag{182}$$

consistent with definition (38). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (182) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{183}$$

Thus, $\Lambda_{\rm hypo}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants $K_{\rm geom}$, $2k_{\rm reg}$, and $\frac{1}{2}$. These arise from the slow–fast coupling through $\mathcal{V}_{\rm skew}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\rm hypo}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\mu_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\rm eff} + \alpha_*\,k_P(\mu_*), \tag{184}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\,\tilde{k}_P(\mu_*), \tag{185}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\mu_*) + \tilde{k}_P(\mu_*)\big). \tag{186}$$

Here E_{11} contains the full effective coefficient $b_{2,\rm eff} = b_2 + \frac{a_2}{2}(1+\lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\rm eff}$ arises in the fast block from the

bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (132)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{187}$$

Since $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ (as $\lambda_R \geq 0$), we have $E_{11} \geq E_{22}$, so the maximum is attained by the first row. To obtain a uniform closed-form upper bound, we additionally replace k_P with $\tilde{k}_P$ (since $\tilde{k}_P \geq k_P$) in E_{11}:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\left[\tfrac{1}{2}\,\tilde{k}_P(\mu_*) + \tfrac{3}{2}\,\tilde{k}_P(\mu_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\mu_*). \tag{188}$$

Recall from Lemma 3.8 that the bounds k_P and $\tilde{k}_P$ depend on $\mu_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of μ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + \frac{3}{4}(a_1 + a_2). \tag{189}$$

Indeed, $2\tilde{k}_P(\mu_*) = 2\big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\mu_*^{-1/2}$; using $\frac{1}{2}k_P + \frac{3}{2}\tilde{k}_P = (2(b_1+b_2)k_{\mathcal{A}} + \frac{3(a_1+a_2)}{4})\mu^{-1/2} = C_{\text{sing}}\mu^{-1/2}$ (tighter than $2\tilde{k}_P$), the bound (188) refines to

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \mu_*^{-1/2}. \tag{190}$$

Substituting $\alpha_*(\gamma)$ from (37), the bound becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{191}$$

consistent with definition (39).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which defines the threshold

$$\eta^*(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|} \geq \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)}. \tag{192}$$

Substituting the explicit forms of Λ_{hypo} (38) and B_{pert} (39) yields

$$\eta^*(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}\big)\big(2b_{2,\text{eff}} + 2\alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}\big)}. \tag{193}$$

Asymptotic analysis of stability. We analyze the scaling of $\eta^*(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.

Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta^*(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta^*(\gamma) \asymp \mathcal{O}(1)$.

Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{194}$$

The singularity thus cancels, and at the optimal friction the stability threshold is well-behaved:

$$\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{195}$$

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter $\delta(\gamma)$ defined by

$$\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\mu_*}\|. \tag{196}$$

Using $\|\mathcal{A}_{\mu_*}\| \le k_{\mathcal{A}}\,\mu_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}$, we obtain the crude but uniform estimate

$$\delta(\gamma) \le 2\,\frac{\gamma\lambda_R}{Z(\gamma)^2}\,k_{\mathcal{A}}\,\mu_*^{-1/2} = \frac{\gamma\sqrt{\lambda_R}\,\mu_*^{-1/2}}{\big(\gamma\Theta\mu_*^{-1/2}+K_{\mathrm{geom}}\big)^2}. \tag{197}$$

Setting $x := \gamma\Theta\mu_*^{-1/2}$ gives

$$\delta(\gamma) \le \frac{\sqrt{\lambda_R}}{\Theta}\,\frac{x}{(x+K_{\mathrm{geom}})^2} \le \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\mathrm{geom}}} =: \delta_{\max}. \tag{198}$$

Since $K_{\mathrm{geom}} \ge \sqrt{M_R/\lambda_R}$ and $\Theta = \sqrt{M_R}/2$, we obtain $\delta_{\max} \le \frac{\lambda_R}{2M_R}$. By Assumption 3.1.3, the flux conditioning constant M_R is monotone (any larger value also satisfies the assumption); we may therefore take $M_R \ge \lambda_R$ without loss of generality, which gives $\delta_{\max} \le \frac{1}{2}$. (Choosing $M_R > \lambda_R$ can only *increase* $\delta_{\max} = \lambda_R/(2M_R)$ relative to its natural value $\lambda_R/(2M_R^\natural)$ where $M_R^\natural$ is the smallest valid constant; when $M_R^\natural \ge \lambda_R$ already, no enlargement is needed. The assumption $M_R \ge \lambda_R$ is verified in both examples of Section 5.) This implies that the condition number of the norm equivalence is uniformly bounded for all γ:

$$C_\gamma := \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}} \le \sqrt{\frac{1+1/2}{1-1/2}} \tag{199}$$

$$= \sqrt{3}. \tag{200}$$

Thus, we have the uniform equivalence bounds:

$$\frac{1}{4}\|f\|^2 \le \frac{1}{2}(1-\delta(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\|f\|^2 \le \frac{3}{4}\|f\|^2. \tag{201}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{202}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{203}$$

Next we use the norm equivalence from Proposition 3.10 in the convenient form

$$\frac{1}{2}(1-\delta(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\,\|f\|^2, \tag{204}$$

where $\delta(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\mu_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f). \tag{205}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t). \tag{206}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\mu_*}(f_0). \tag{207}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \frac{1+\delta(\gamma)}{1-\delta(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}t}\,\|f_0\|^2. \tag{208}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta(\gamma)}. \tag{209}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{210}$$

For the perturbation norm, we use the sharp bound (190) at $\mu_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{211}$$

Substituting these into the expression for λ_η and using the uniform bound $1+\delta(\gamma) \le 3/2$:

$$\lambda_\eta \ge \frac{2}{3}\left[\frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}} - \eta\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\right)\right]. \tag{212}$$

This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/Z(\gamma)^2$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}}f\|_{\mathcal{H}} \le \sqrt{3}\,e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{213}$$

This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine γ_{opt} maximising $\Lambda_{\text{hypo}}(\gamma)$. From (182), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)}\cdot\frac{x}{(x+K_{\text{geom}})^2 + \left(2k_{\text{reg}} + \frac{1}{2}\right)}. \tag{214}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\text{geom}})^2 + C_0]$ with $C_0 := 2k_{\text{reg}} + \frac{1}{2}$ is maximised when $(x+K_{\text{geom}})^2 + C_0 = 2x(x+K_{\text{geom}})$, equivalently $x^2 = K_{\text{geom}}^2 + C_0$. Translating back:

$$\gamma_{\text{opt}} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\text{lift}}\sqrt{\lambda(Q_O)}, \tag{215}$$

where $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}$.

Maximal rate. At $\gamma = \gamma_{\text{opt}}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\text{opt}} = \sqrt{K_{\text{geom}}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\text{geom}})^2 + C_0 = 2(K_{\text{geom}}^2 + C_0 + K_{\text{geom}}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\text{geom}})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{216}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Stability threshold at optimal lift. The admissible perturbation range $\eta^*(\gamma_{\rm opt})$ at the accelerated regime is determined as follows. From (192), the threshold depends on $\|\mathbf{E}(\gamma)\|$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at $\gamma_{\rm opt}$:

$$\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2 + 2k_{\rm reg}}. \tag{217}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2 + 2k_{\rm reg} + \frac{1}{2}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2 + 2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (190):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{218}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $\|\mathbf{E}(\gamma_{\rm opt})\| = \mathcal{O}(1)$, the stability threshold scales as

$$\eta^*(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2\|\mathbf{E}(\gamma_{\rm opt})\|} \asymp \sqrt{\lambda(Q_O)}. \tag{219}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 4.4.

D.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{220}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{221}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le C\, s(\mathcal{L}). \tag{222}$$

Proof. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt. \tag{223}$$

The assumed exponential bound $\|e^{t\mathcal{L}} f\| \le Ce^{-\nu t}\|f\|$ implies that the Bochner integral converges in $\mathcal{H}_0$ and

$$\|u\| \le \int_0^\infty \|e^{t\mathcal{L}} f\|\, dt \le \int_0^\infty Ce^{-\nu t}\, dt\, \|f\| = \frac{C}{\nu}\, \|f\|. \tag{224}$$

Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L}e^{t\mathcal{L}} f\, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big)\, dt = f. \tag{225}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le C/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/C$, i.e. $\nu \le C\, s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume there exist constants* $C \geq 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{226}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \geq 0\,;\ \|e^{t\mathcal{L}} f\| \leq e^{-1}\, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{227}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2C\, s(\mathcal{L})}. \tag{228}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [4]*.*

Proof. By the assumed exponential bound $\|e^{t\mathcal{L}}\| \leq Ce^{-\nu t}$, for any $r \geq 0$ we have $\|e^{r\mathcal{L}}\| \leq C$. By the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for all $t \geq 0$ we can write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}_0$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$, and obtain from the semigroup property and $\|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\| \leq e^{-1}$:

$$\|e^{t\mathcal{L}}\| \leq \|e^{r\mathcal{L}}\|\, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \leq C\, e^{-n}.$$

As in the proof of Lemma D.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}}\, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \leq \int_0^\infty \|e^{t\mathcal{L}}\|\, dt \leq \sum_{n=0}^{\infty} \int_0^{t_{\mathrm{rel}}(\mathcal{L})} C\, e^{-n}\, dr = \frac{C\, t_{\mathrm{rel}}(\mathcal{L})}{1 - e^{-1}} \leq 2C\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \geq (2C\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 4.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{229}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi \in \mathcal{H}_F, \tag{230}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \geq 1. \tag{231}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = P_F \mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{232}$$

Consequently,

$$\mathcal{L} f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{233}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \ +\ \eta\mathcal{V}_{\mathrm{pert}} f_\gamma. \tag{234}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}}\psi + P_F \mathcal{V}_{\mathrm{skew}}\psi. \tag{235}$$

For the slow part, using $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$ and $P_S\phi = \phi$, we have

$$P_S \mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{236}$$

For the fast part, apply the flux regularity estimate (11) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\rm skew}P_S\phi = \mathcal{V}_{\rm skew}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\rm skew}\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\rm ell}\,\|(-Q_O)\phi\| = \kappa_{\rm ell}\,\lambda(Q_O). \tag{237}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\rm skew}\psi\| \le (1+\kappa_{\rm ell})\,\lambda(Q_O). \tag{238}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\rm pert}f_\gamma\|. \tag{239}$$

To bound the perturbation, use Assumption 3.1.4 to estimate

$$\|\mathcal{V}_{\rm pert}f\| \le \|\mathcal{K}_{\rm pert}f\| + \|\mathcal{S}_{\rm pert}f\| \tag{240}$$

$$\le (a_1+a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1+b_2)\,\|f\|, \qquad f\in{\rm dom}((-\mathcal{R})^{1/2}). \tag{241}$$

Since $\phi\in\ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{242}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi,\psi\rangle = \langle\mathcal{V}_{\rm skew}\phi,(-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi,\phi\rangle = \lambda(Q_O). \tag{243}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert}f_\gamma\| \le (b_1+b_2)\,\|f_\gamma\| + (a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{244}$$

Dividing by $\|f_\gamma\|\ge 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{245}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 5.5

We verify Assumption 3.1.1–4 for the forced Langevin decomposition on the core $\mathcal{C}_{\rm Lan} := C^\infty(\mathbb{T}^d)\otimes\mathcal{S}(\mathbb{R}^d)\subset L^2(\mu_0)$.

E.1. **Microscopic spectral gap and centering.** The Ornstein–Uhlenbeck operator $\mathcal{R} = -p\cdot\nabla_p + \beta^{-1}\Delta_p$ acts only on p-variables and has spectral gap $\lambda_R = 1$ on $\mathcal{H}_F = \ker(\Pi_p)^\perp$ (classical; see e.g. [26]). For Assumption 3.1.2: $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ (Hamiltonian flow preserves μ_0), $P_S\mathcal{V}_{\rm skew}P_S = 0$ (Gaussian centering $\int p\,d\nu_p = 0$), and $P_S\mathcal{V}_{\rm pert}P_S = 0$ (since $\nabla_p\phi(q) = 0$).

E.2. **Flux regularity (Assumption 3.1.3).** For $\psi = \psi(q)\in\mathcal{H}_S$, the flux is $g_\psi = p\cdot\nabla_q\psi\in\mathcal{H}_F$. Since $\mathcal{R}(p_i) = -p_i$, we have $(-\mathcal{R}^+)g_\psi = g_\psi$, giving $M_R = 1$.

For the elliptic regularity bound, define $Z_{ij}(p) := p_ip_j - \beta^{-1}\delta_{ij}$. A direct computation yields

$$P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi = \sum_{i,j} Z_{ij}(p)\,\partial_{q_iq_j}\psi. \tag{246}$$

By Cauchy–Schwarz and Gaussian moment calculations ($\mathbb{E}[p_i^4] = 3\beta^{-2}$),

$$\|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi\|_{L^2(\mu_0)} \le c_{d,\beta}\,\|\nabla_q^2\psi\|_{L^2(\nu)}, \qquad c_{d,\beta} = \frac{\sqrt{d(d+1)}}{\beta}. \tag{247}$$

It remains to control $\|\nabla_q^2\psi\|_{L^2(\nu)}$ by $\|(-Q_O)\psi\|_{L^2(\nu)}$.

Step 1 (Hessian vs. Laplacian). By Parseval's identity on $\mathbb{T}^d$, $\|\nabla_q^2\psi\|_{L^2(dq)} = \|\Delta_q\psi\|_{L^2(dq)}$ for periodic ψ. Switching to $L^2(\nu)$ with $\rho_{\min} \le \rho \le \rho_{\max}$ ($\rho \propto e^{-\beta U}$) gives

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le \sqrt{\rho_{\max}/\rho_{\min}}\,\|\Delta_q\psi\|_{L^2(\nu)}. \tag{248}$$

Step 2 (Laplacian vs. Q_O). From $\Delta_q\psi = \beta(Q_O\psi + \nabla U\cdot\nabla_q\psi)$ and the Dirichlet-form identity $\langle(-Q_O)\psi,\psi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q\psi\|^2_{L^2(\nu)}$, one derives via Cauchy–Schwarz and the Poincaré inequality for ν:

$$\|\nabla_q\psi\|_{L^2(\nu)} \le \beta\sqrt{C_{\mathrm{P}}(\nu)}\,\|(-Q_O)\psi\|_{L^2(\nu)}, \tag{249}$$

and hence $\|\Delta_q\psi\|_{L^2(\nu)} \le \beta(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)})\,\|(-Q_O)\psi\|_{L^2(\nu)}$, where $M_U := \|\nabla U\|_{L^\infty}$.

Combining Steps 1–2 with (247) yields the flux regularity bound $\kappa_{\mathrm{ell}} = c_{d,\beta}\cdot\sqrt{\rho_{\max}/\rho_{\min}}\cdot\beta(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)})$.

E.3. **Perturbation energy bounds (Assumption 3.1.4).** By integration by parts in p, the $L^2(\mu_0)$-adjoint is $\mathcal{V}^\dagger_{\mathrm{pert}} = -F\cdot\nabla_p + \beta(F\cdot p)$, giving the skew/symmetric decomposition:

$$\mathcal{K}_{\mathrm{pert}} = F\cdot\nabla_p - \tfrac{\beta}{2}(F\cdot p), \qquad \mathcal{S}_{\mathrm{pert}} = \tfrac{\beta}{2}(F\cdot p). \tag{250}$$

Since $\|\nabla_p f\| \le \sqrt{\beta}\,\|(-\mathcal{R})^{1/2}f\|$, the bounds reduce to controlling the multiplier $\|pf\|$ by $\|f\|$ and $\|\nabla_p f\|$. A Gaussian integration-by-parts argument with Young's inequality gives the explicit p-multiplier bound:

$$\||p|f\|_{L^2(\mu_0)} \le \sqrt{2d/\beta}\,\|f\|_{L^2(\mu_0)} + (2/\beta)\,\|\nabla_p f\|_{L^2(\mu_0)}. \tag{251}$$

Combining with $\|F\|_{L^\infty} < \infty$ yields Assumption 3.1.4(a)–(b) with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$. □

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